

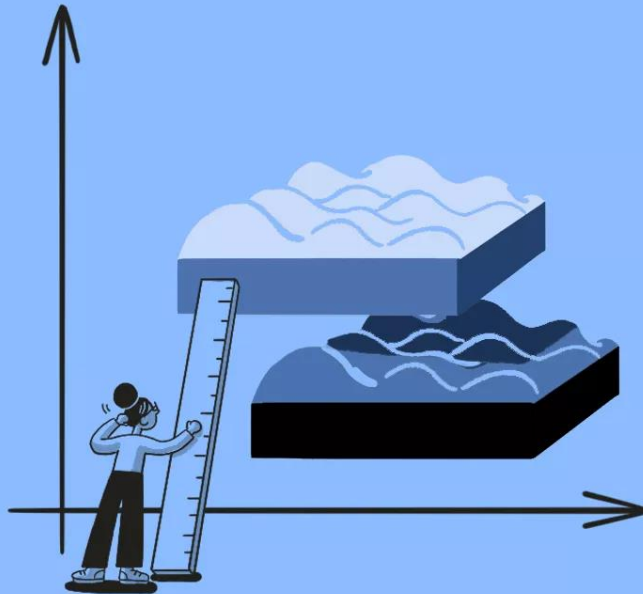
Lecture 13 – Time Series III

Autocorrelation

Autocorrelation

[,ò-tō-,kòr-ə-'lā-shən]

A mathematical representation of the degree of similarity between a given time series and a lagged version of itself over successive time intervals.



Today's Learning Outcomes

1. Understand what autoregressive process is
2. Understand ACF and PACF
3. Understand stationarity

Dynamic stochastic processes (lags)

In last lectures, we discussed deterministic trends. However, history can have stochastic effects, inducing serial correlation in our data

Several conceptually different ways to think of the stochastic effect of history

1. Past realizations, such as y_{t-1} , influence current levels of y
2. Past shocks to y , such as y_{t-1} , influence current levels of y
3. Past expectations of y , such as $\mathbb{E}(y_{t-1})$, influence current levels of y

Dynamic stochastic processes (lags)

To focus on these more subtle dynamics, we will do two things for the rest of this topic:

- We will work in time series which we can assume have been *detrended*; that is, assume we have removed $t\hat{\beta}$ if necessary
- We will ignore any other covariates that might influence y

Example of dynamic stochastic processes

Past realizations of y influence current levels of y .

Examples: today's air temperature is affected by yesterday's air temperature

Past shocks to y influence current levels of y

Examples: the response of today's streamflow to the rainfall events yesterday?

Past expectations of y influence current levels of y

Examples: Groundwater levels responding with a lag to past climate inputs

Let's incorporate these dynamics into our baseline model

Past realizations of y

$$y_t = y_{t-1}\phi_1 + \varepsilon_t$$

Known as an autoregressive process

Each new realization of y_t incorporates the last period's realization, y_{t-1}

Note that only one lag of y_t appears in our model.

This is an AR(1) process, or an autoregressive process of degree 1.

However, the distant past still has an effect. Implied by above:

$$y_{t-1} = y_{t-2}\phi_1 + \varepsilon_{t-1}$$

and so

$$y_{t-2} = y_{t-3}\phi_1 + \varepsilon_{t-2}$$

... and on and on back to the “original” period

Autoregressive Processes

Recursive reparameterization

Iterating through all past periods and substituting back into the first formula.

For AR(1), recursion reveals the following:

$$\begin{aligned}y_t &= y_{t-1}\phi_1 + \varepsilon_t \\&= (y_{t-2}\phi_1 + \varepsilon_{t-1})\phi_1 + \varepsilon_t \\&= y_{t-2}\phi_1^2 + \varepsilon_{t-1}\phi_1 + \varepsilon_t \\&= (y_{t-3}\phi_1 + \varepsilon_{t-2})\phi_1^2 + \varepsilon_{t-1}\phi_1 + \varepsilon_t \\&= y_{t-3}\phi_1^3 + \varepsilon_{t-2}\phi_1^2 + \varepsilon_{t-1}\phi_1 + \varepsilon_t \\&\dots \text{ substitute through } y_{t-k} \\&= y_{t-k}\phi_1^k + \sum_{j=0}^{k-1} \varepsilon_{t-j}\phi_1^j \\&\dots \text{ substitute through } y_{t-\infty} \\&= \sum_{j=0}^{\infty} \varepsilon_{t-j}\phi_1^j\end{aligned}$$

Autoregressive Processes

$$y_t = \sum_{j=0}^{\infty} \varepsilon_{t-j} \phi^j$$

In the limit, if y_t is AR(1), then

y_t includes the effects of every disturbance back to the beginning of time: $\varepsilon_{t-1}, \dots, \varepsilon_{t-\infty}$

Effects of history lasts forever in the autoregressive processes

So what would happen if $|\phi_1| < 1$?

And if $|\phi_1| > 1$?

And if $|\phi_1| = 1$ exactly?

Autoregressive Processes

If $-1 < \phi_1 < 1$, the effect of the past approaches zero as time passes, but never completely fades

If $|\phi_1| > 1$, the process is explosive, tending quickly to infinity.
Not a reasonable model of anything in the long-run,
and few earth system processes in the short run

if $|\phi_1| = 1$ exactly, we have a random walk or unit root.
Very persistent effects of history, and many unusual properties.

Simulating time series

Basic idea:

1. Start at some y_0 , e.g., $y_0 = 0$, and choose a ϕ_1 to study
2. Draw $y_1 = y_0\phi_1 + \varepsilon_1$, where $\varepsilon_1 \sim N(0, \sigma^2)$
3. Iterate on step 2 to generate, y_2, y_3 , etc.

Simulating time series

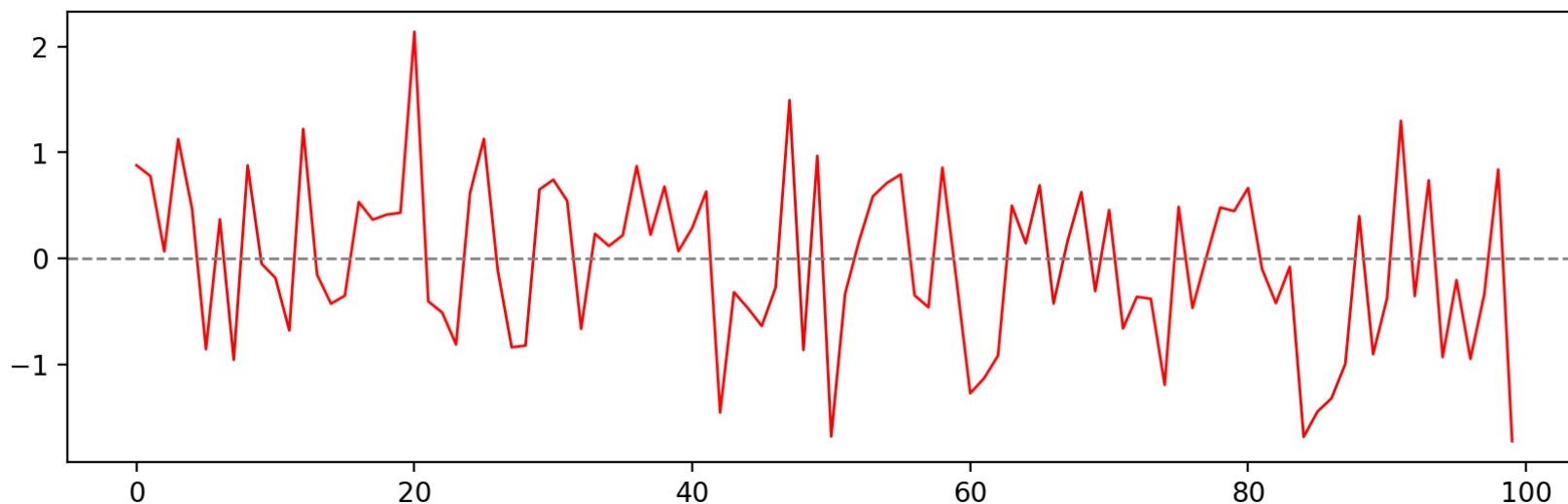
A more flexible simulator (available in CourseMaterials)

AR(1) process with $\phi_1 = 0.6$

```
y = make_ts( n = 100,  
            ar = [0],  
            trend = 0,  
            ma = None,  
            seasons = [0]*12,  
            adjust = "level",  
  
            varY = 1,  
            initY = 0,  
            initE = 0,  
            burnin = 10,  
            random_state = 42)
```

```
# Simulate 100 periods  
# AR(1):  $\phi_1=0.6$   
# Determ. trend (+0.01 unit/period)  
# MA(0)  
# 12 period cycle  
# If "factor", multiply  $y(t)$  by the  
#     seasonal value; if "level", add it.  
# Error term variance  
# Prior level of time series, for lags  
# Prior error, for lags  
# Discard this many iterations at start
```

Simulated AR(1) process with $\phi_1 = 0$



Here are 100 draws from an AR(1) with $\phi_1 = 0$

Nice, but can we “see” that these iterations are not serially correlated (i.e., a “white noise” process, or just $N(0, \sigma^2)$)?

Review: Correlation coefficients

Simplest way to “see” autocorrelation is to calculate and plot the correlation between observations separated by a given number of periods j , where $j = 1, 2, 3, \dots$

But it helps to first review what exactly we mean by correlation. Recall:

$$r_{x,y} = cor(x, y) = \frac{cov(x, y)}{\sqrt{var(x)}\sqrt{var(y)}}$$

This is what Python computes when asked for the correlation of two variables like those on the right

t	y_t	x_t
1	−0.6	3.8
2	9.7	7.2
3	−4.1	3.8
4	7.2	−0.8
5	0.7	−9.6
6	0.9	8.1
7	6	6.2
8	3.5	−2.1
9	−7.2	9.9
10	−2.4	6.7

Review: Correlation coefficients

Simplest way to “see” autocorrelation is to calculate and plot the correlation between observations separated by a given number of periods j , where $j = 1, 2, 3, \dots$

Autocorrelation functions correlate y_t with itself, but lagged j periods,

$$ACF_j = cor(y_t, y_{t-j}) = \frac{cov(y_t, y_{t-j})}{\sqrt{var(y_t)}\sqrt{var(y_{t-j})}}$$

Note that for an AR(1) with ϕ_1 , the ACF_1 should be ϕ_1

t	y_t	y_{t-1}
1	-0.6	NA
2	9.7	-0.6
3	-4.1	9.7
4	7.2	-4.1
5	0.7	7.2
6	0.9	0.7
7	6	0.9
8	3.5	6
9	-7.2	3.5
10	-2.4	-7.2

```
from statsmodels.graphics.tsaplots import plot_acf
from statsmodels.tsa.stattools import acf
```

Review: Correlation coefficients

Simplest way to “see” autocorrelation is to calculate and plot the correlation between observations separated by a given number of periods j , where $j = 1, 2, 3, \dots$

Autocorrelation functions correlate y_t with itself, but lagged j periods,

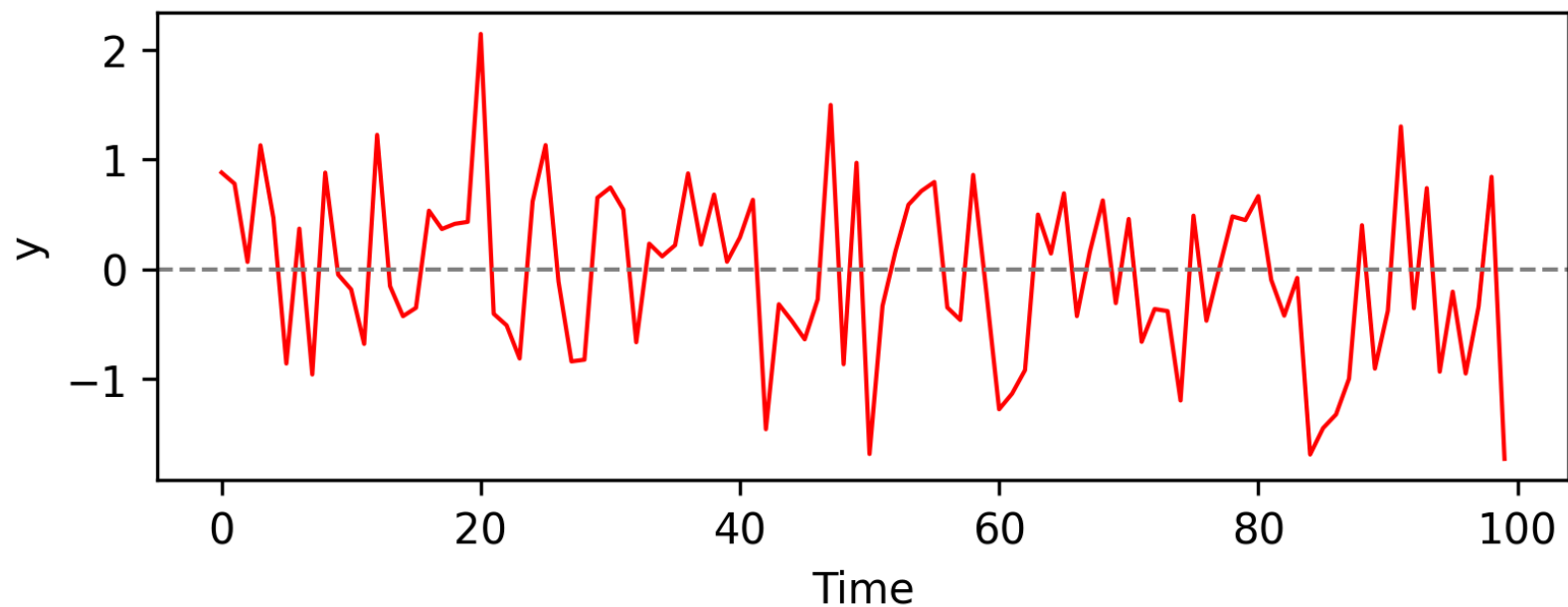
$$ACF_j = cor(y_t, y_{t-j}) = \frac{cov(y_t, y_{t-j})}{var(y)}$$

Note simplification above, and the implicit assumption that $var(y)$ is stable over time

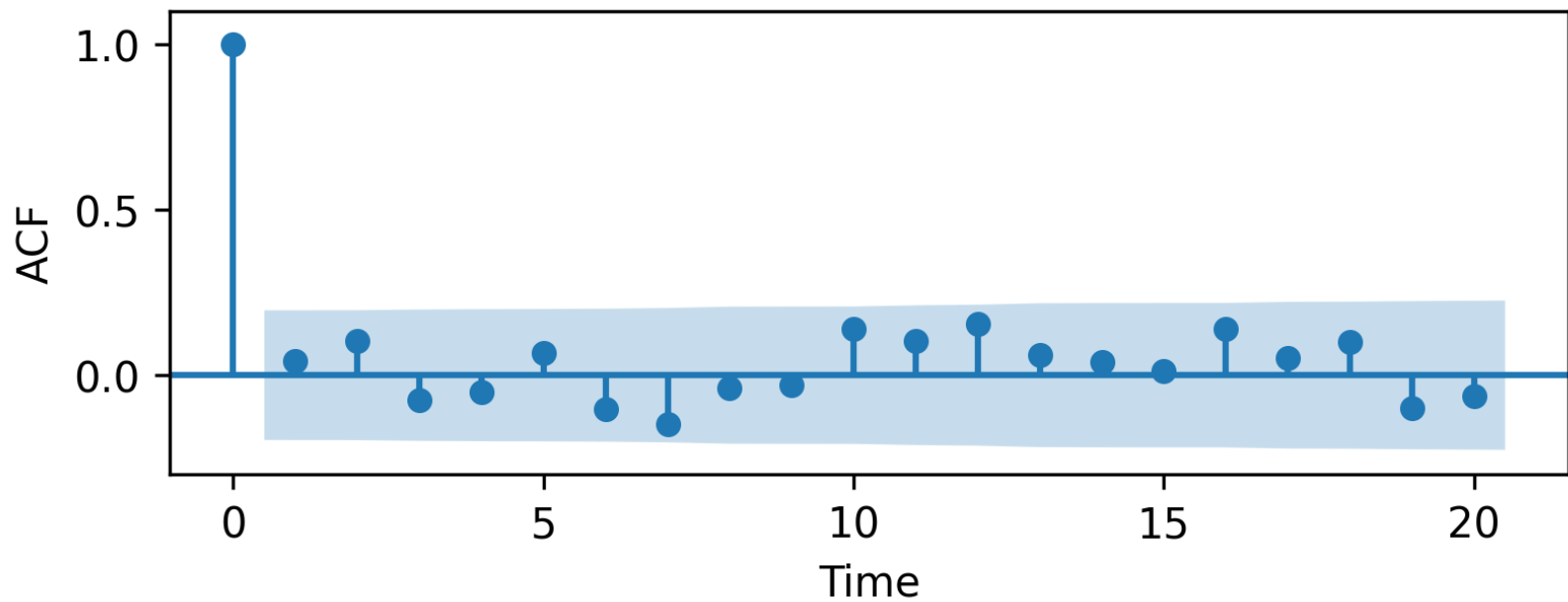
t	y_t	y_{t-1}
1	-0.6	NA
2	9.7	-0.6
3	-4.1	9.7
4	7.2	-4.1
5	0.7	7.2
6	0.9	0.7
7	6	0.9
8	3.5	6
9	-7.2	3.5
10	-2.4	-7.2

```
from statsmodels.graphics.tsaplots import plot_acf
from statsmodels.tsa.stattools import acf
```

Simulated AR(1) process with $\phi_1 = 0$



ACF of AR(1) process with $\phi_1 = 0$



Autocorrelation functions

A useful refinement of the ACF is to “partial out” the effects of intervening lags

That is, we want to isolate the conditional correlation of y_t and y_{t-j} controlling for the values y_{t-1} to y_{t-j+1}

We call this the partial autocorrelation function, or *PACF*

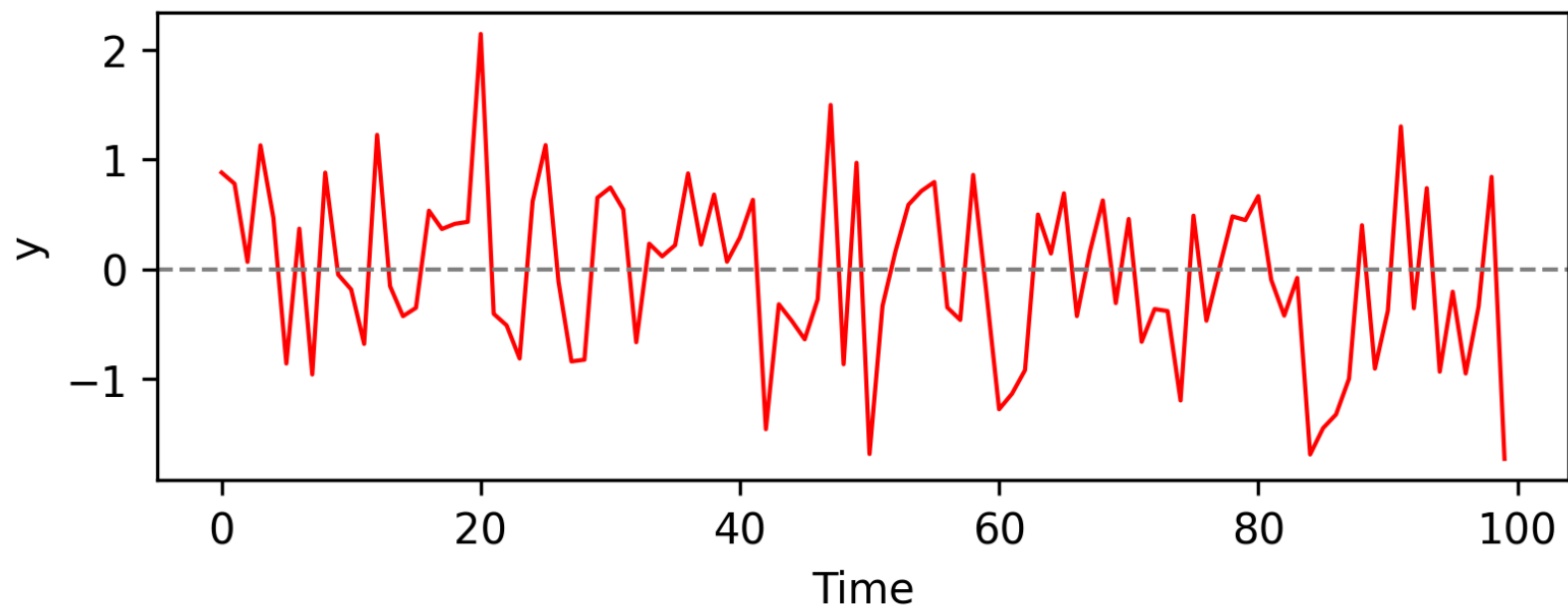
More formally. To compute $PACF_2$, imagine estimating

$$y_t = \beta_0 + \phi_1 y_{t-1} + \varepsilon_t$$

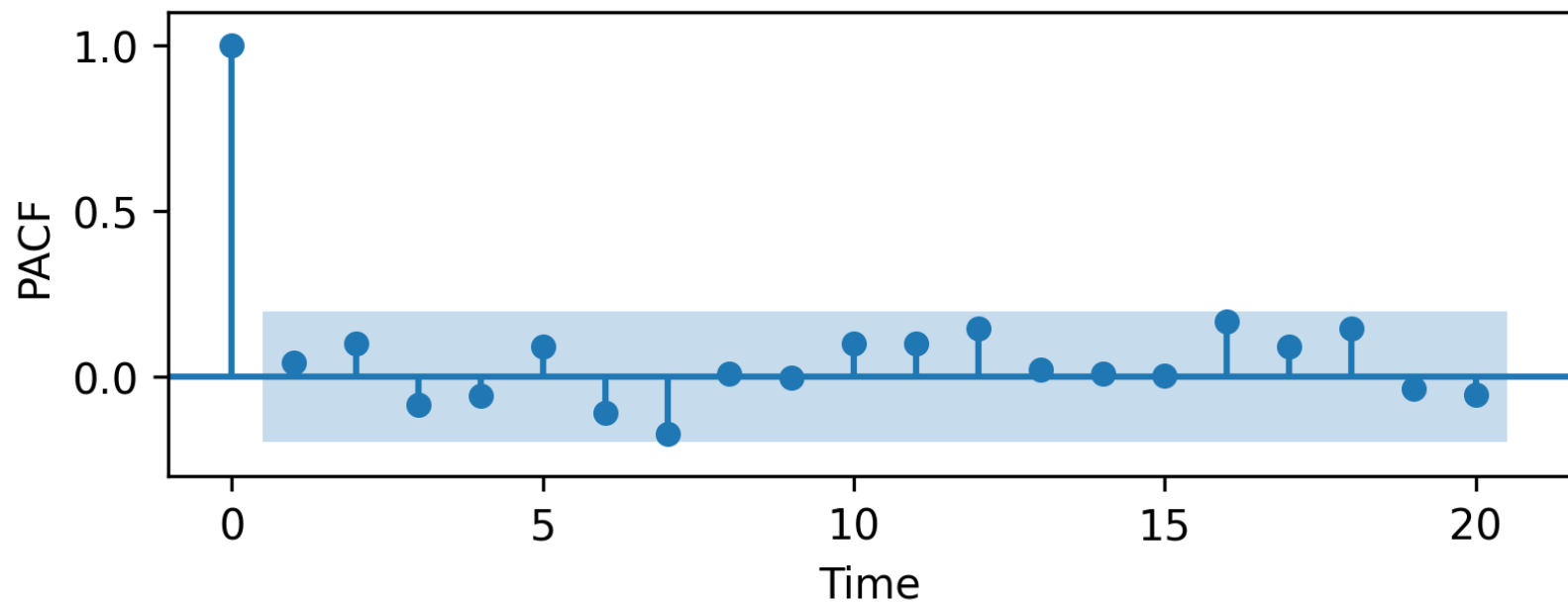
The PACF for $j = 2$ is essentially $cor(y_{t-2}, \hat{\varepsilon}_t)$

What do you expect $PACF_1$ to look like? What does it mean for $PACF_2$ to be zero?

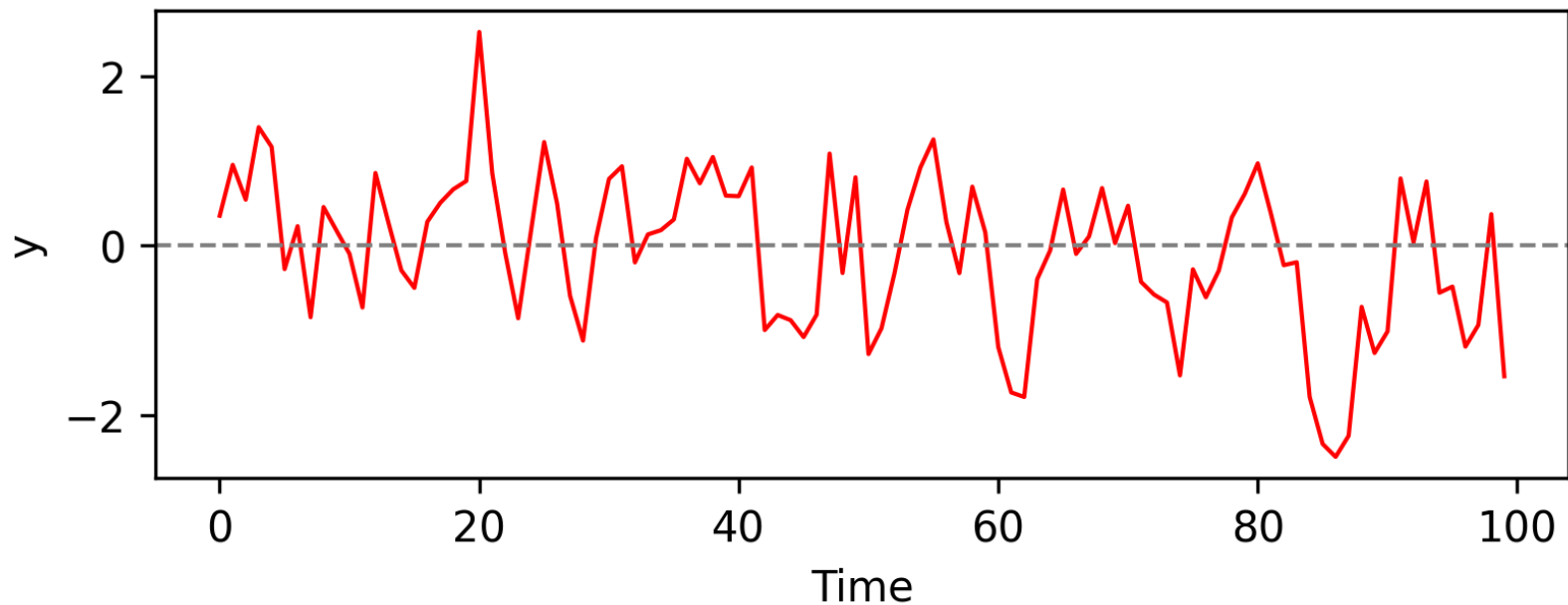
Simulated AR(1) process with $\phi_1 = 0$



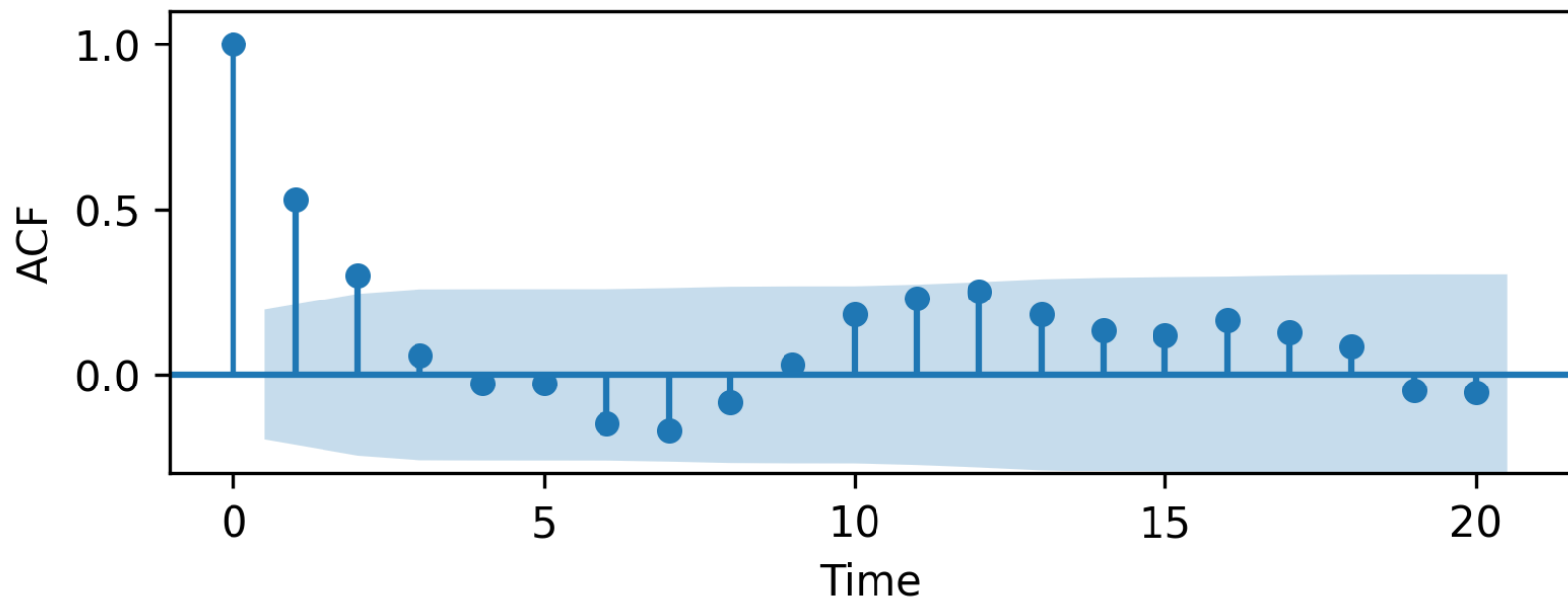
PACF of AR(1) process with $\phi_1 = 0$



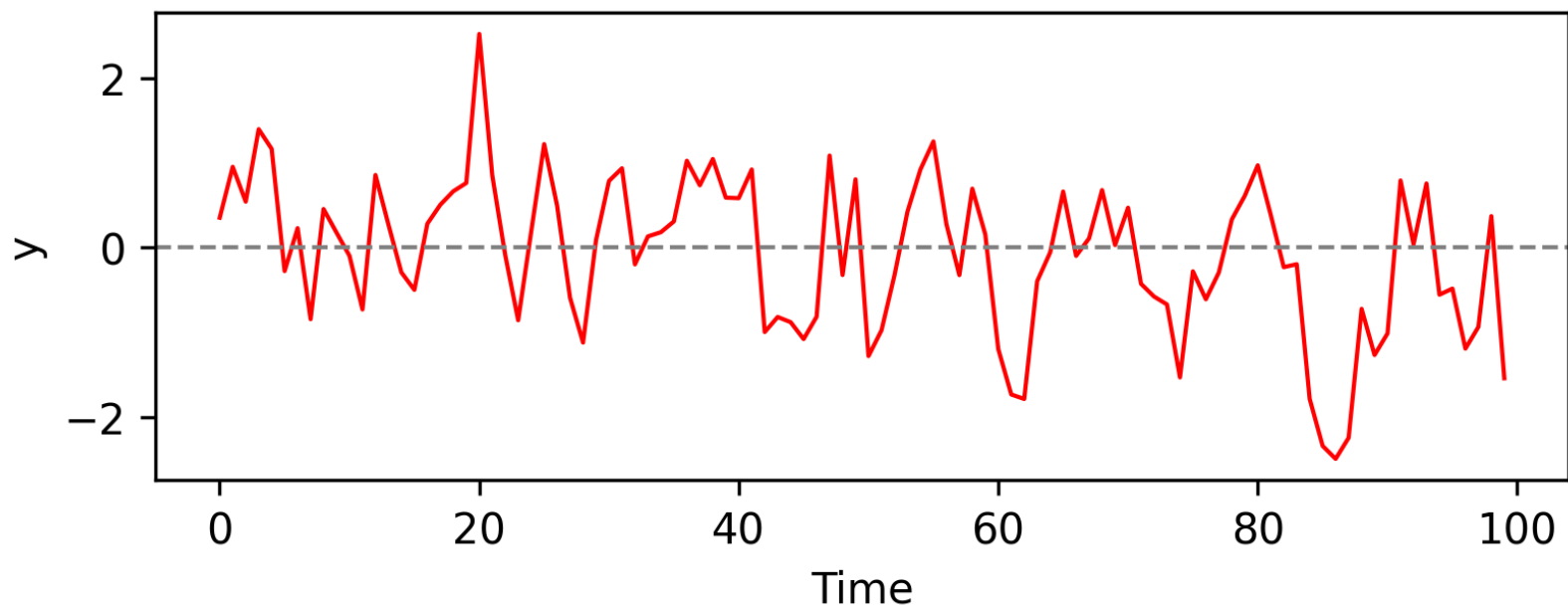
Simulated AR(1) process with $\phi_1 = 0.5$



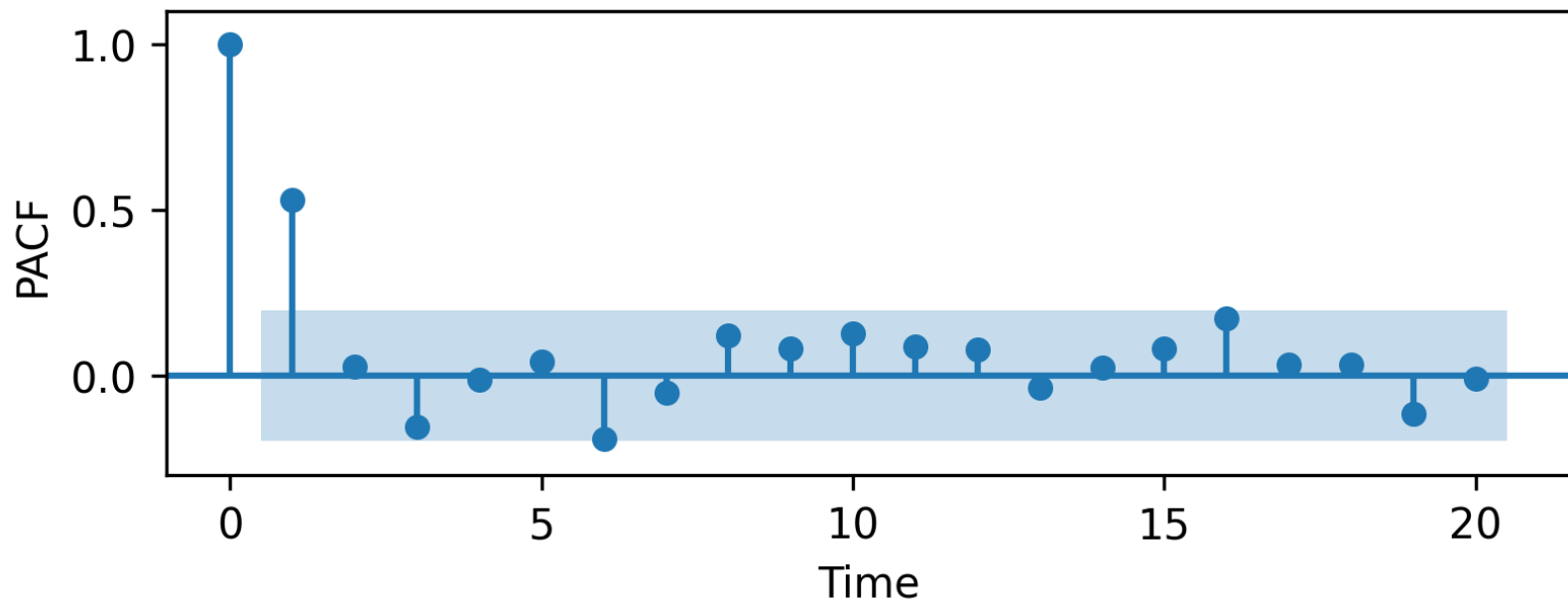
ACF of AR(1) process with $\phi_1 = 0.5$



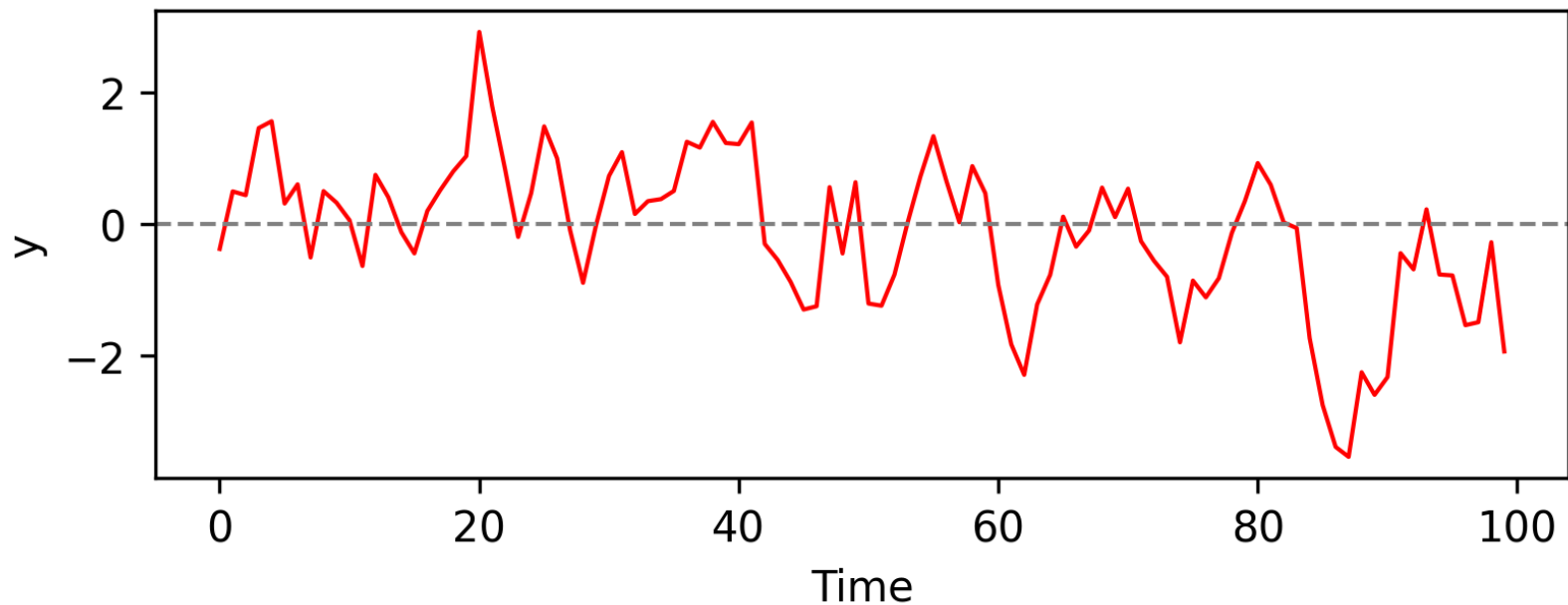
Simulated AR(1) process with $\phi_1 = 0.5$



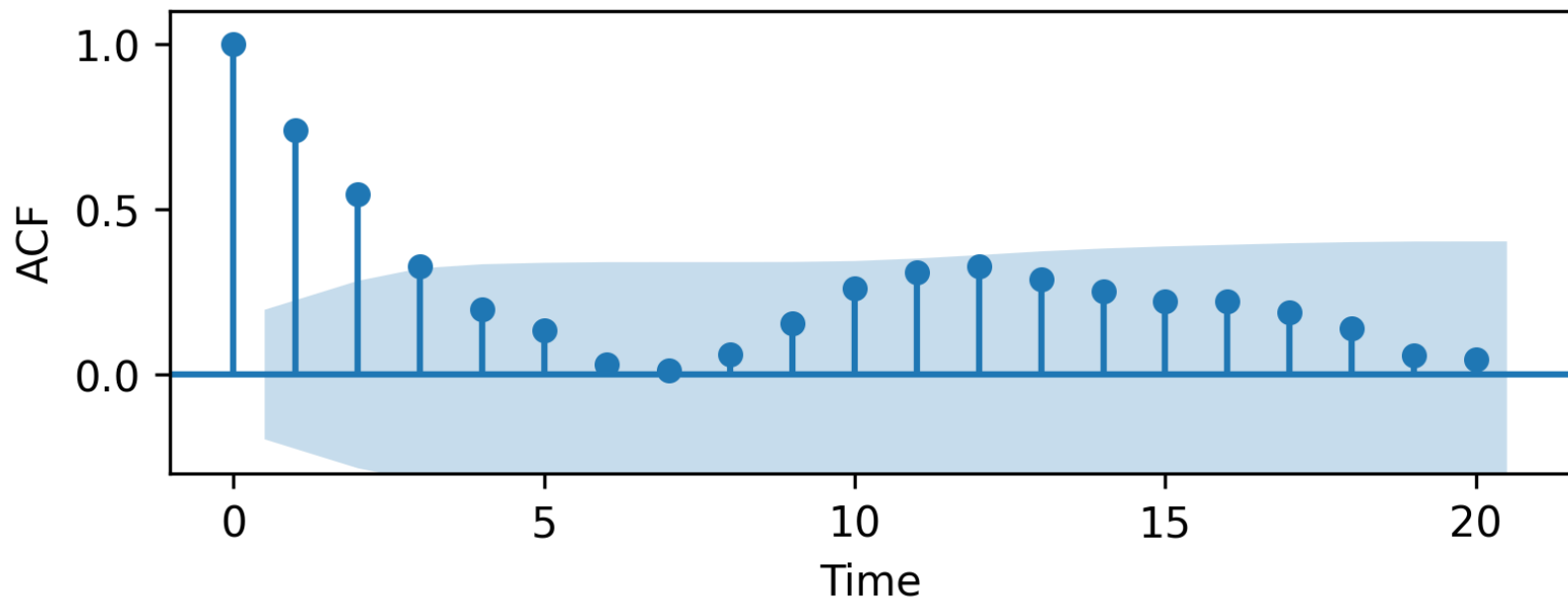
PACF of AR(1) process with $\phi_1 = 0.5$



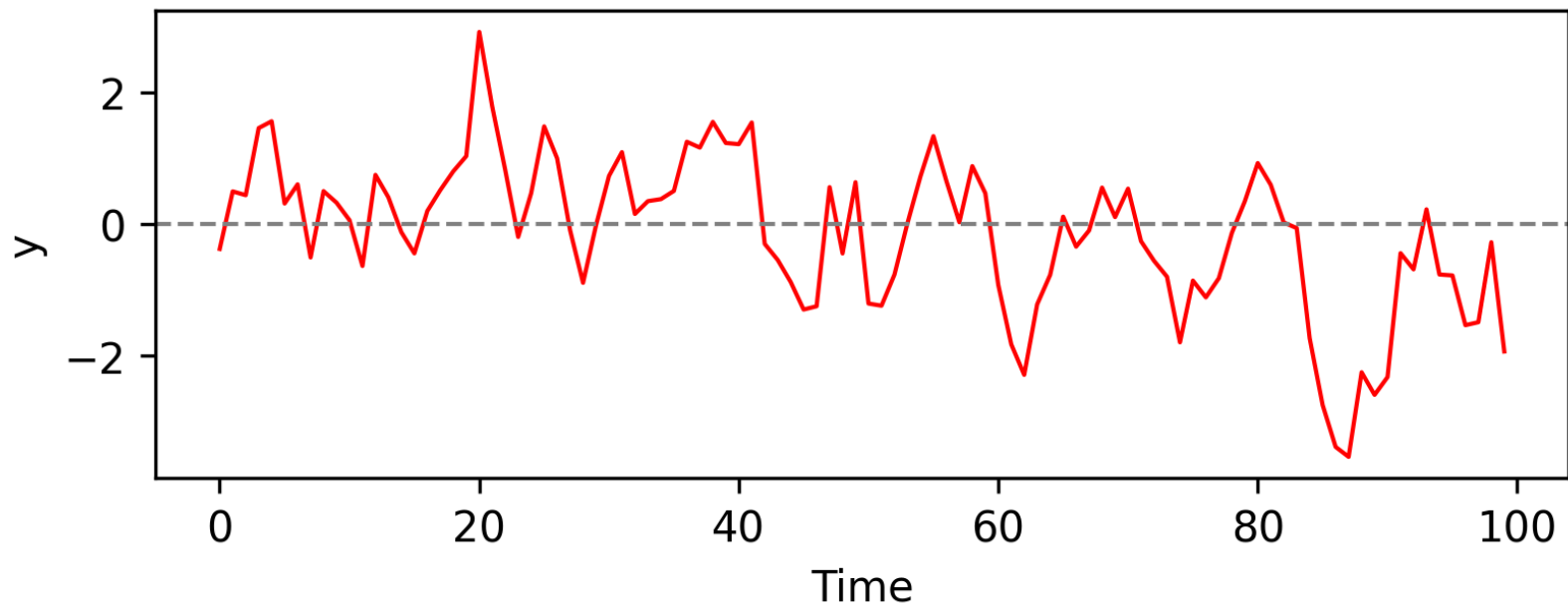
Simulated AR(1) process with $\phi_1 = 0.75$



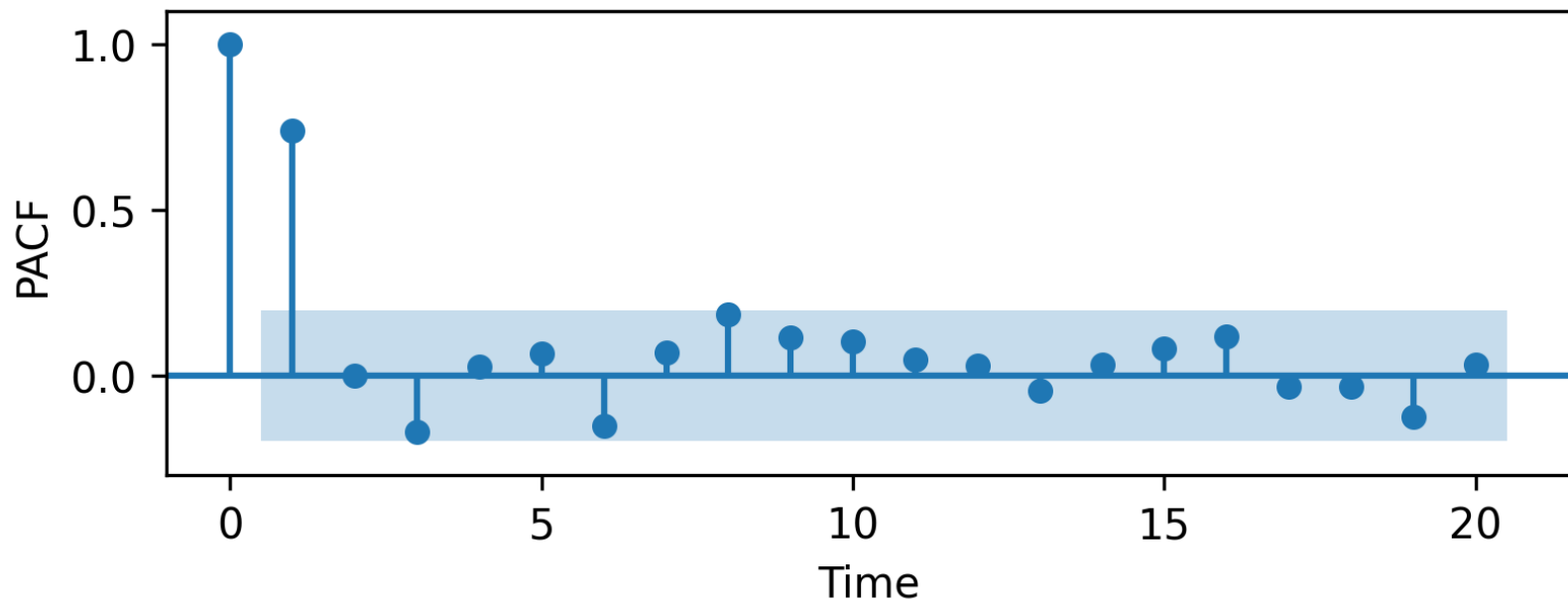
ACF of AR(1) process with $\phi_1 = 0.75$



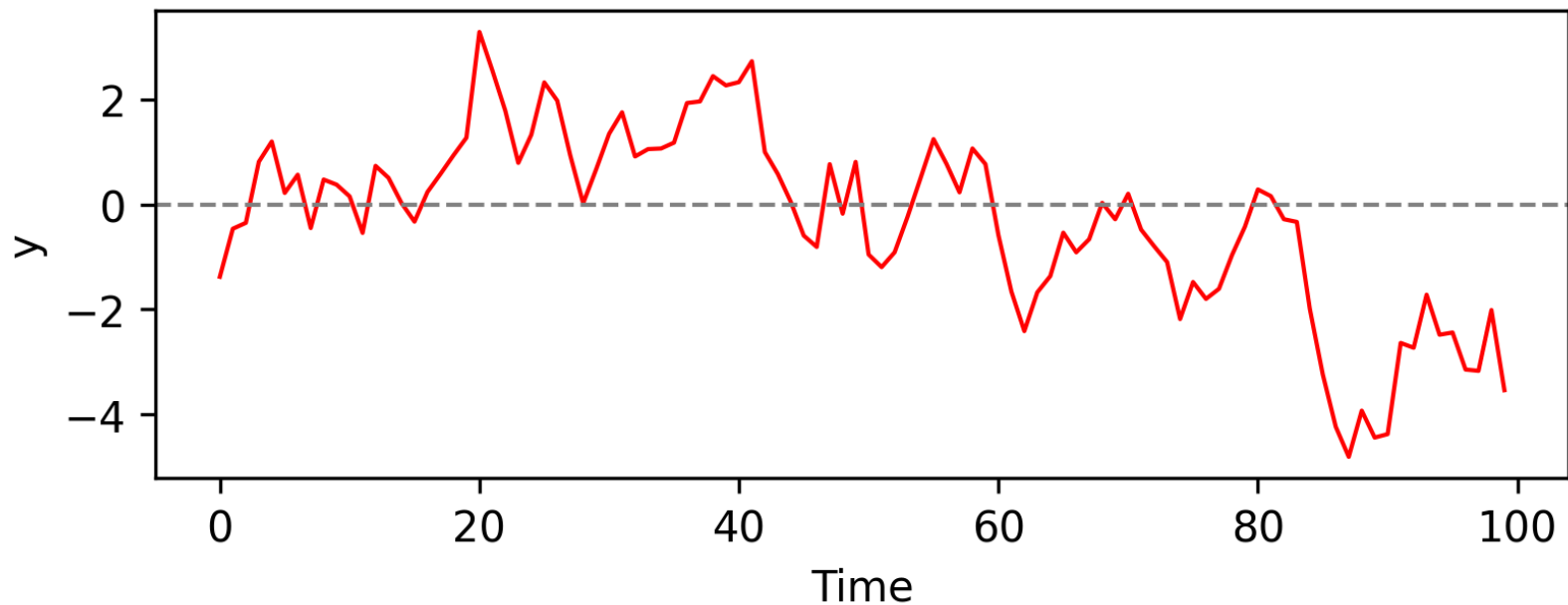
Simulated AR(1) process with $\phi_1 = 0.75$



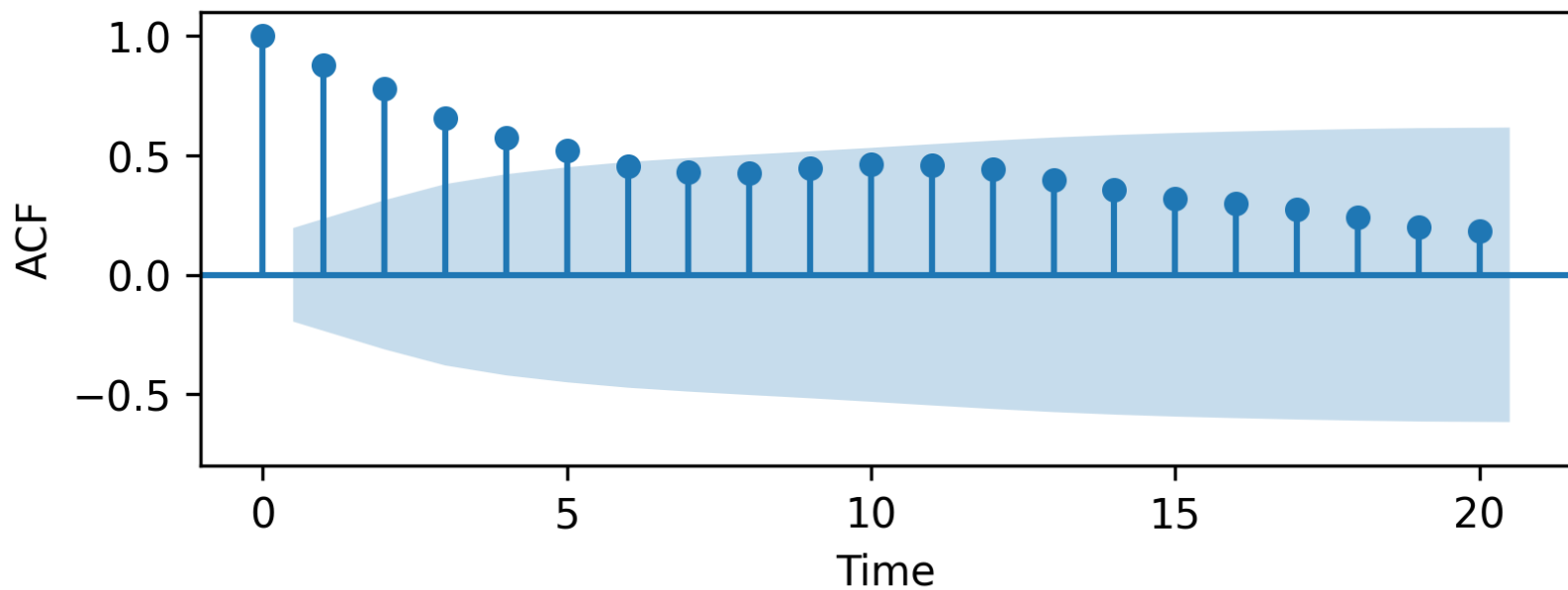
PACF of AR(1) process with $\phi_1 = 0.75$



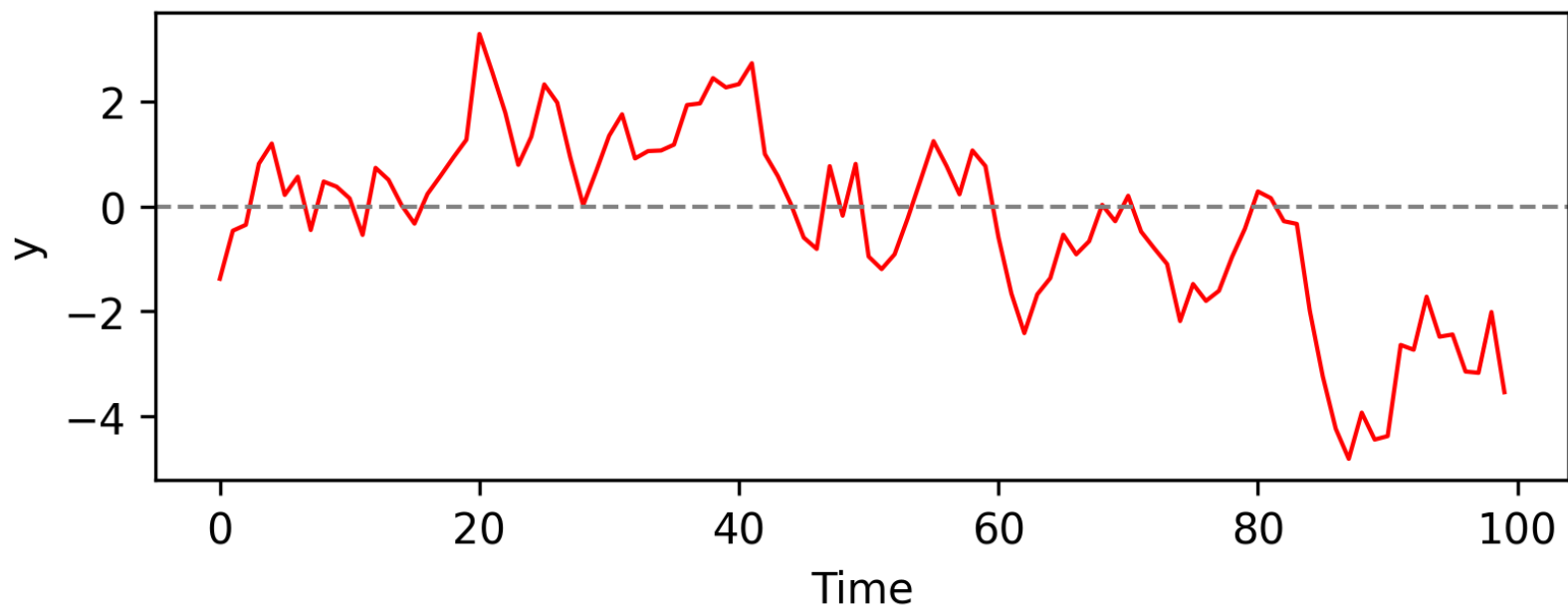
Simulated AR(1) process with $\phi_1 = 0.9$



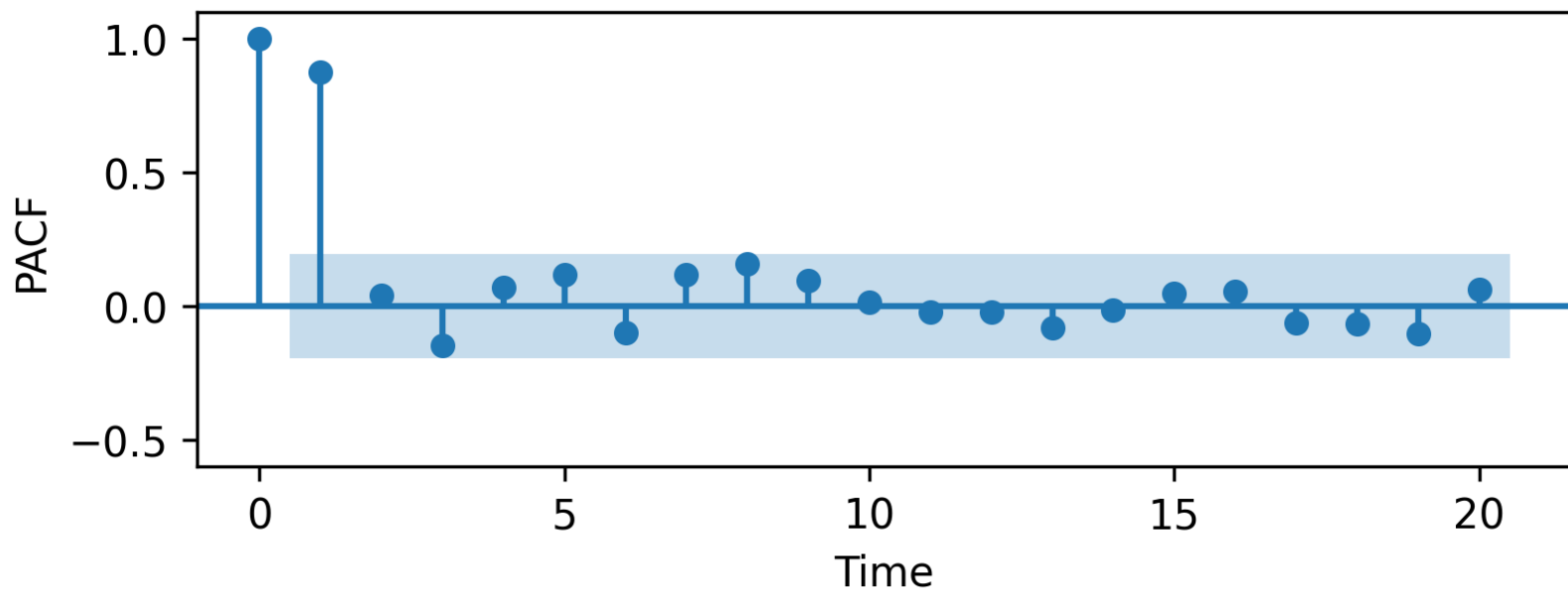
ACF of AR(1) process with $\phi_1 = 0.9$



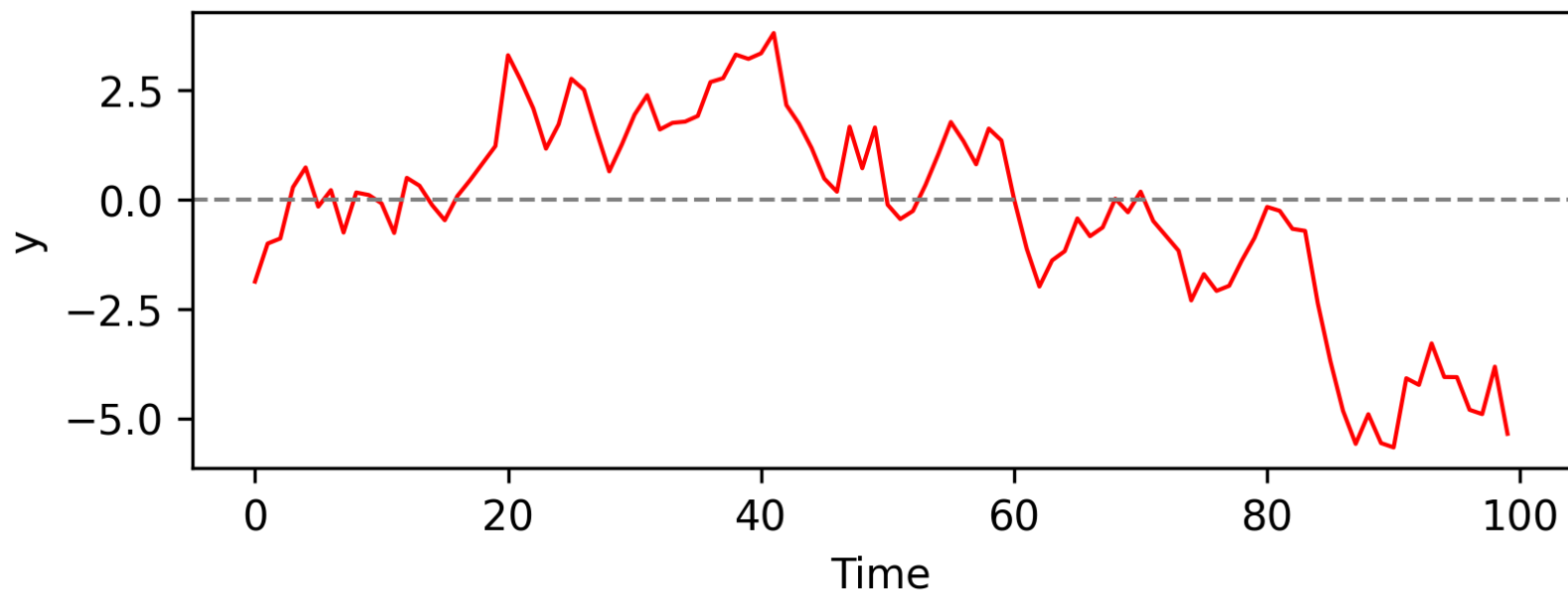
Simulated AR(1) process with $\phi_1 = 0.9$



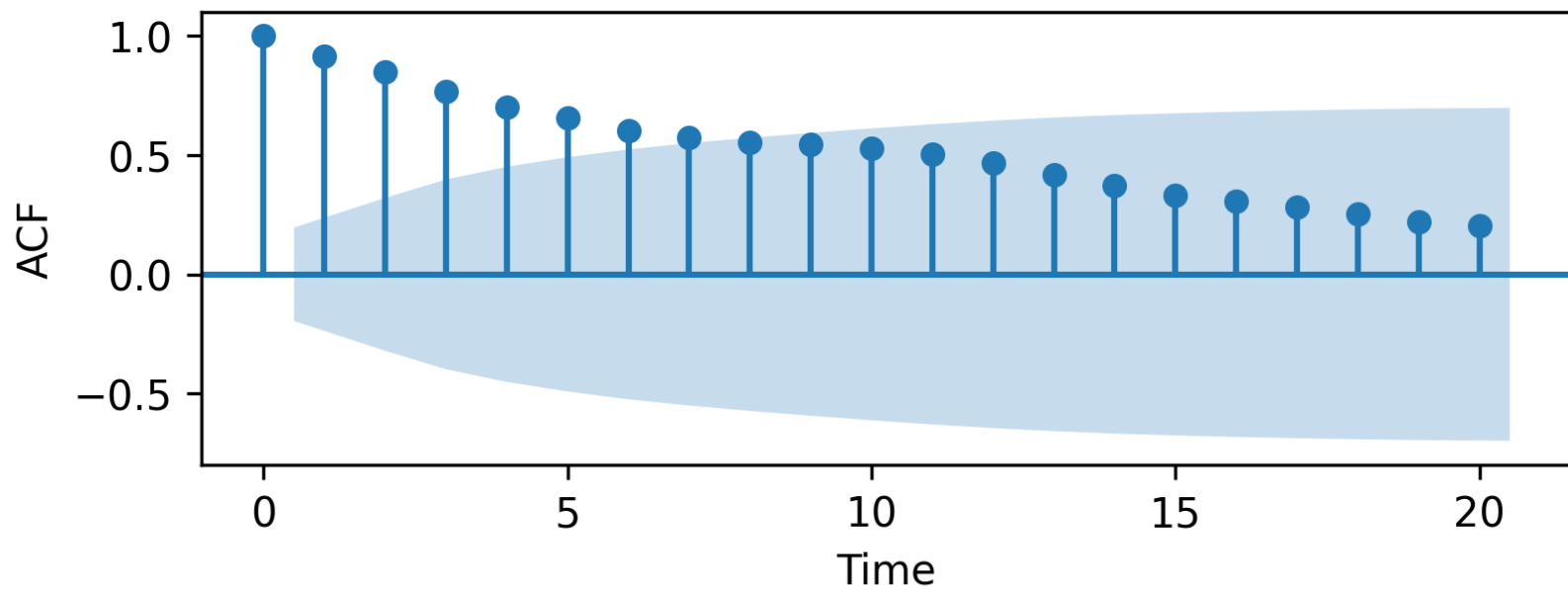
PACF of AR(1) process with $\phi_1 = 0.9$



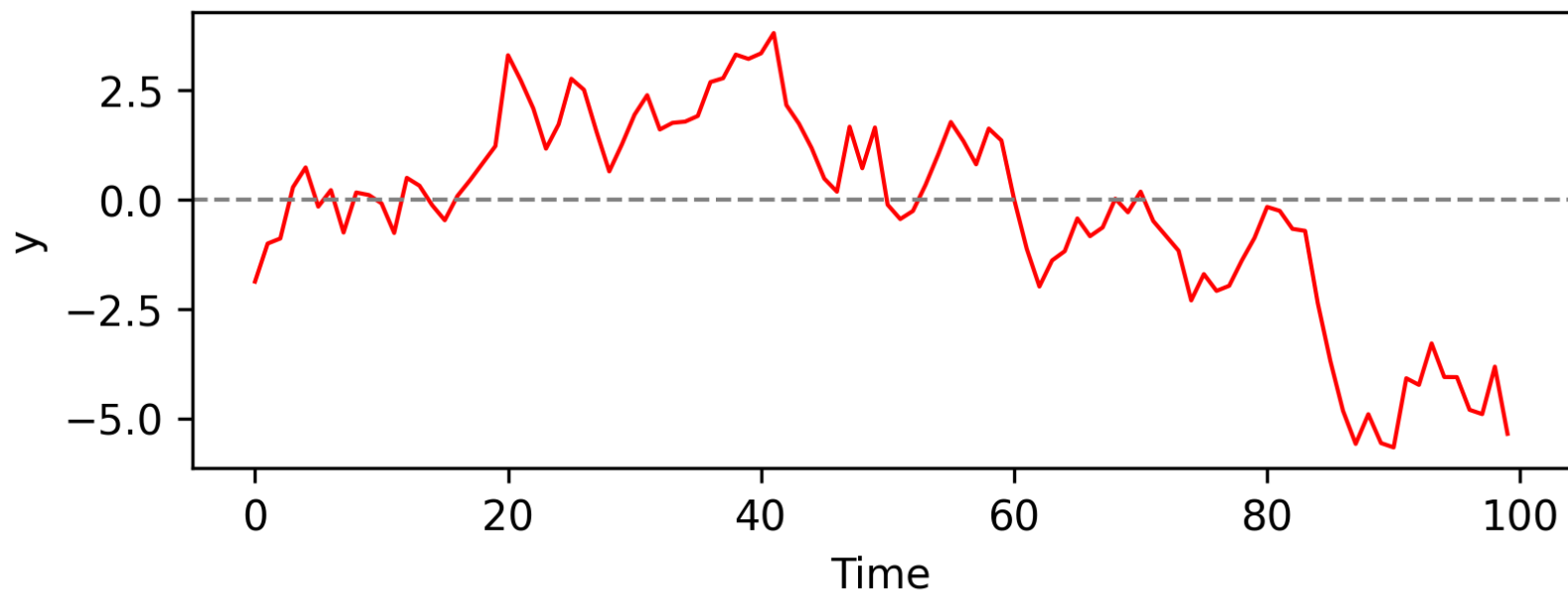
Simulated AR(1) process with $\phi_1 = 0.95$



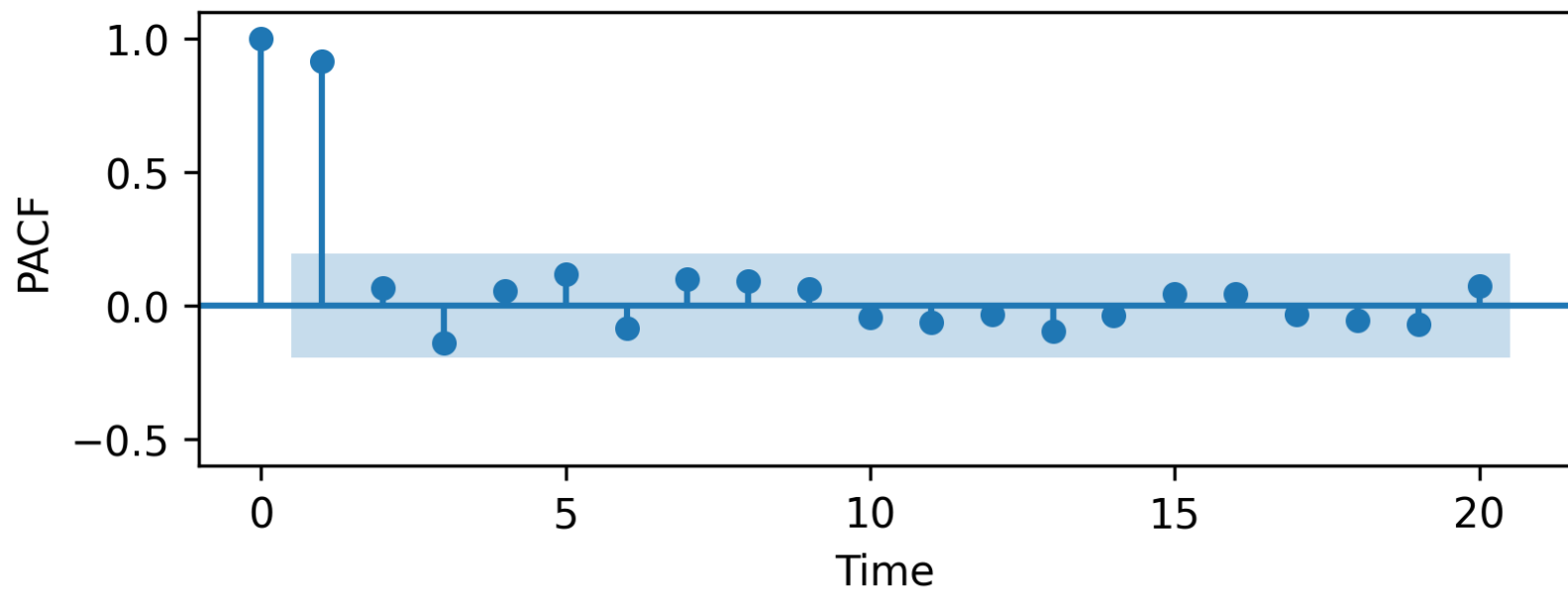
ACF of AR(1) process with $\phi_1 = 0.95$



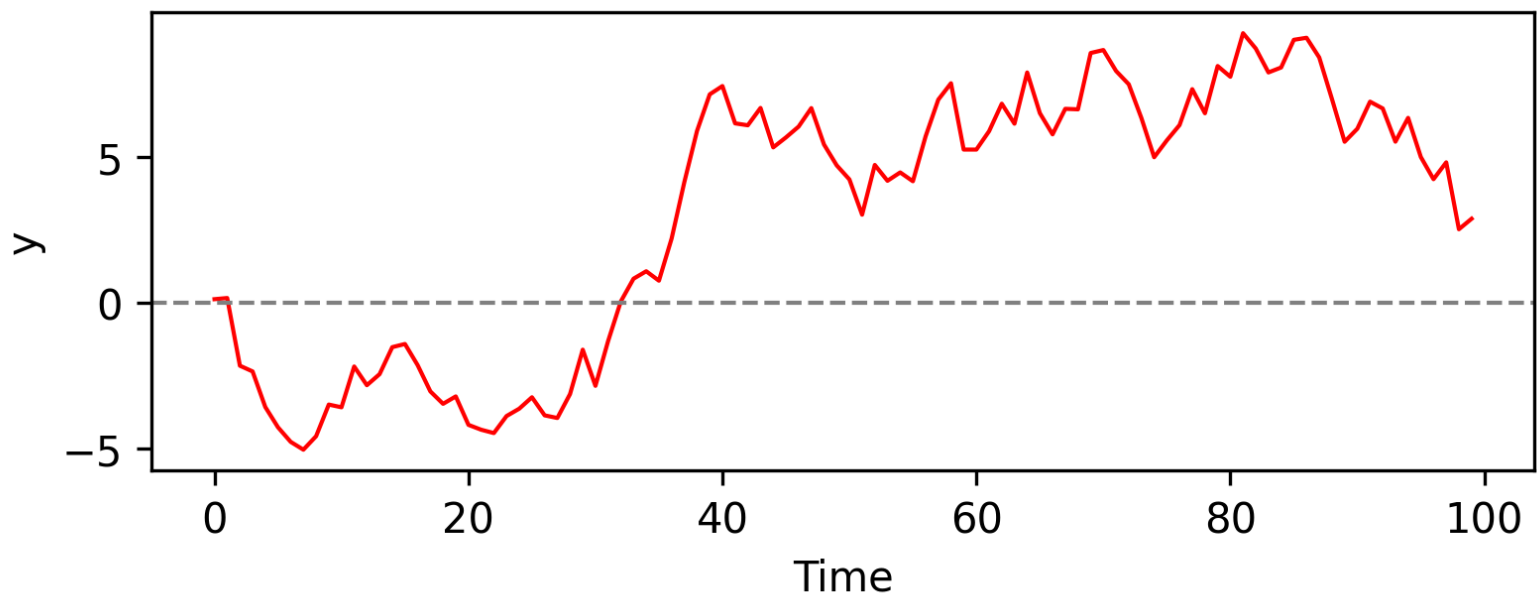
Simulated AR(1) process with $\phi_1 = 0.95$



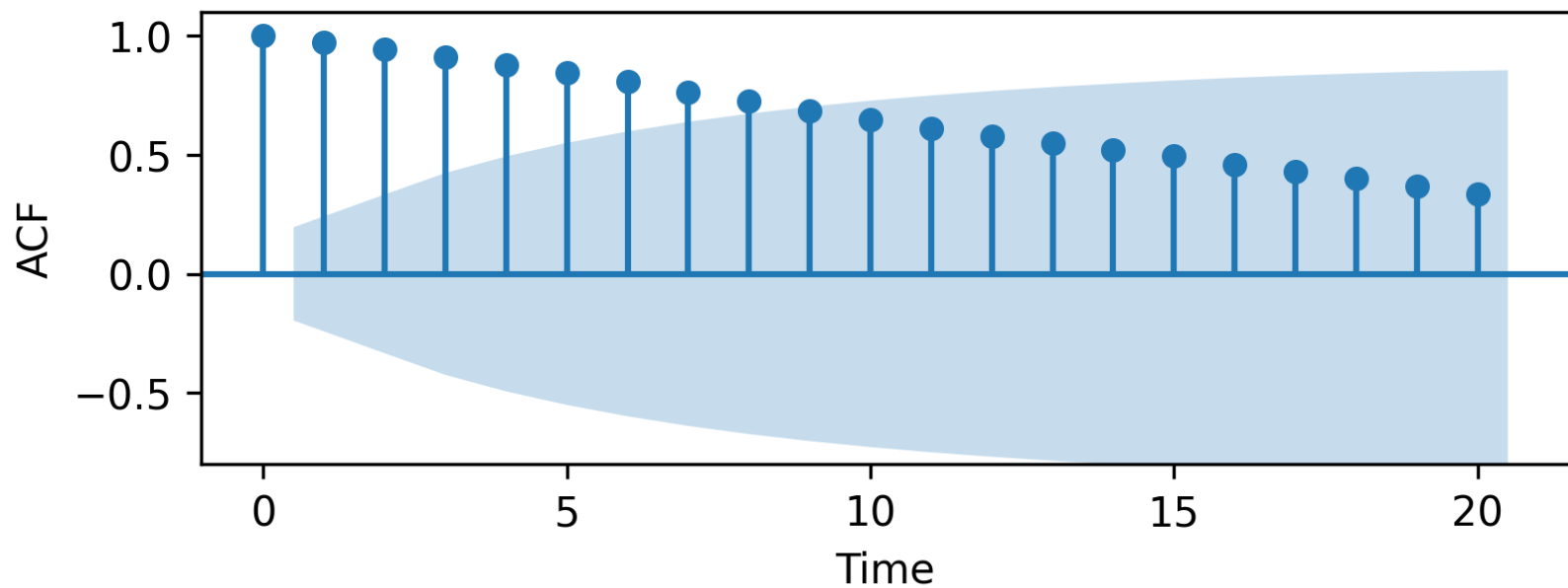
PACF of AR(1) process with $\phi_1 = 0.95$



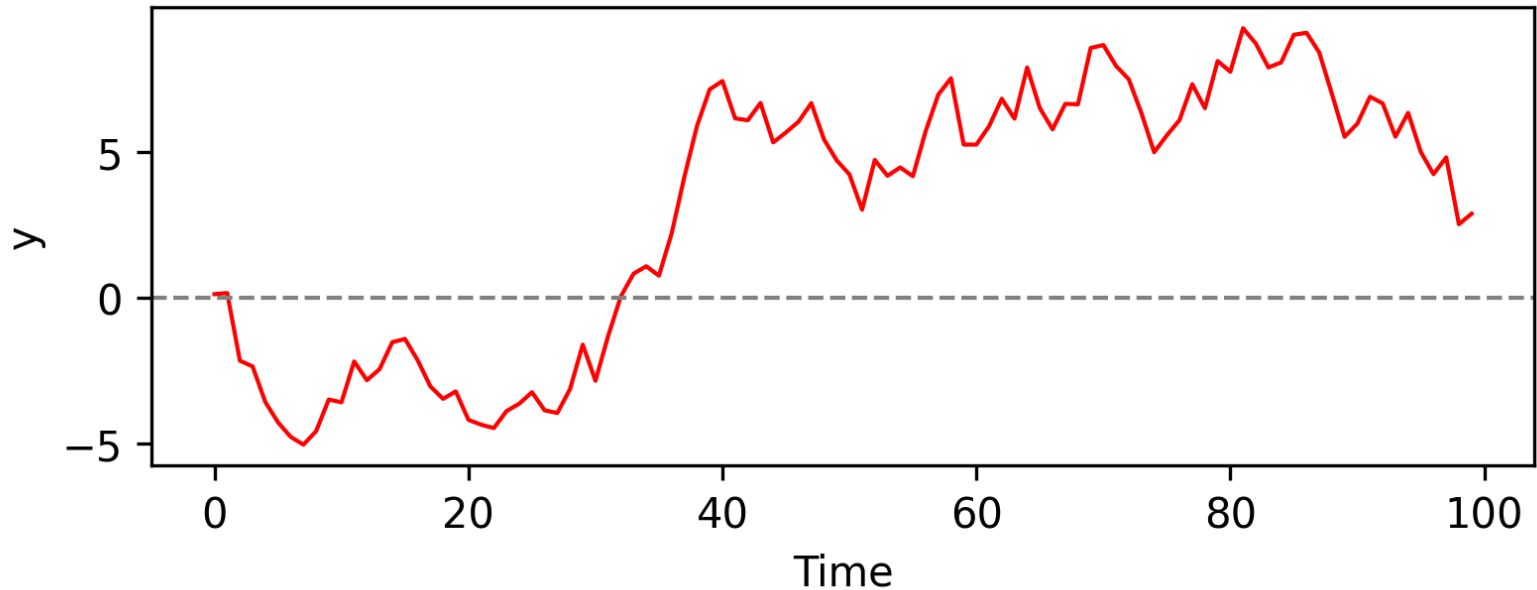
Simulated AR(1) process with $\phi_1 = 0.99$



ACF of AR(1) process with $\phi_1 = 0.99$



Simulated AR(1) process with $\phi_1 = 0.99$

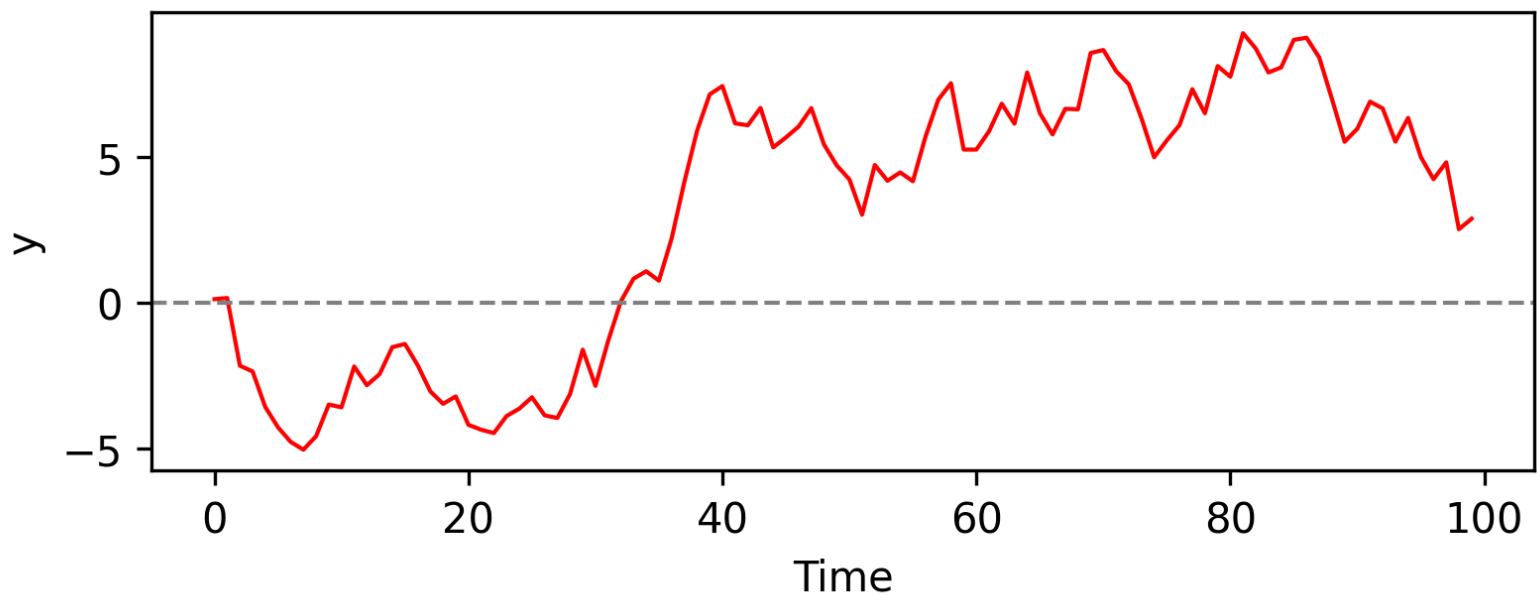


It may seem there is a deterministic trend in this time series

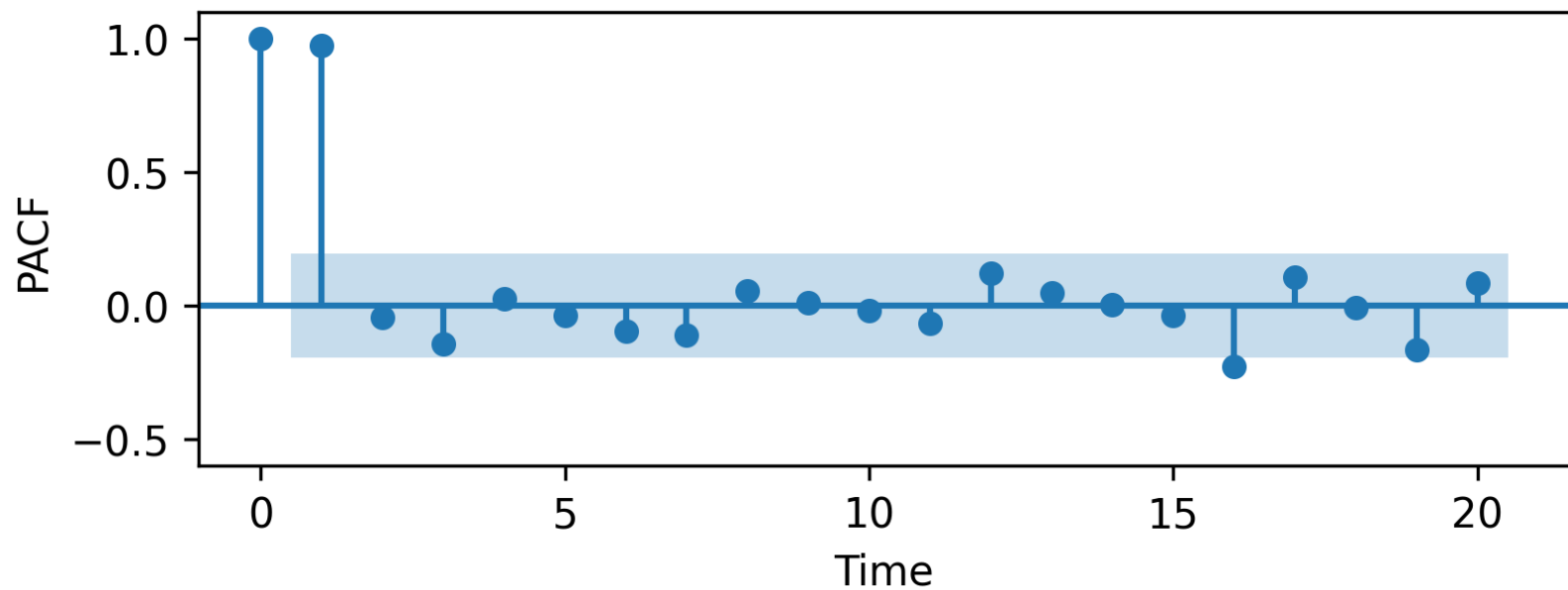
There isn't – the trend above is purely stochastic

Distinguishing whether trends are deterministic or stochastic isn't easy...

Simulated AR(1) process with $\phi_1 = 0.99$

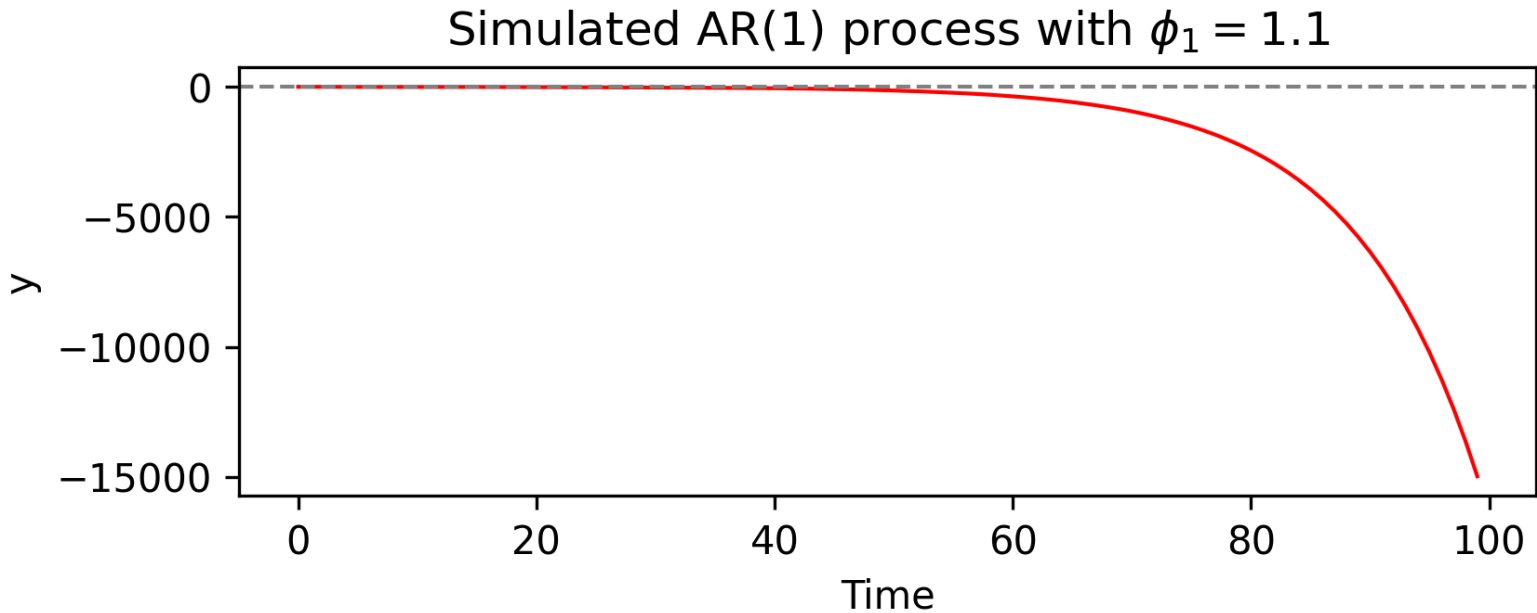


PACF of AR(1) process with $\phi_1 = 0.99$



Observations about AR(1) when $|\phi_1| < 1$

- As $|\phi_1| \rightarrow 0$, series reverts to its mean at 0 quickly
- As $|\phi_1| \rightarrow 1$, series takes longer to revert to mean.
- Still gets there eventually. (Even for $\phi_1 = 0.99$?)
- ACF appears to gradually decline in the lag, as expected
- ACF decays more slowly as $|\phi_1| \rightarrow 1$
- But for all process, PACF is large only for lag of 1, because all series are AR(1)



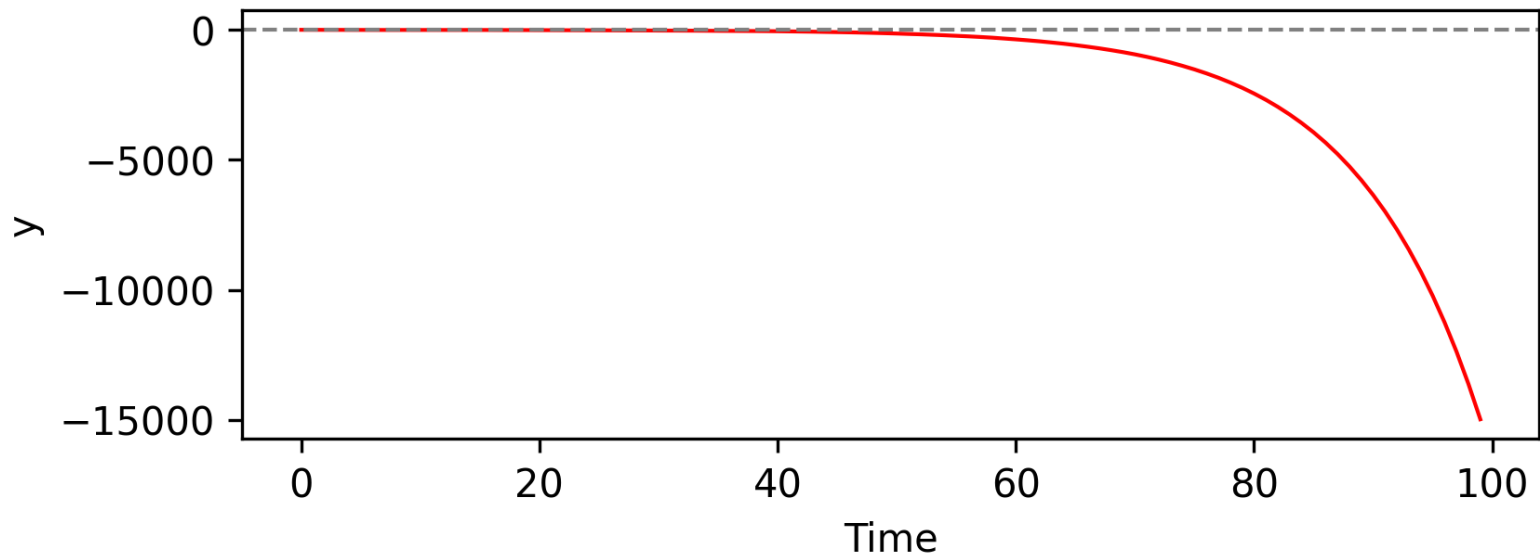
What's happening here?

An explosive process strongly influenced by an initial event.

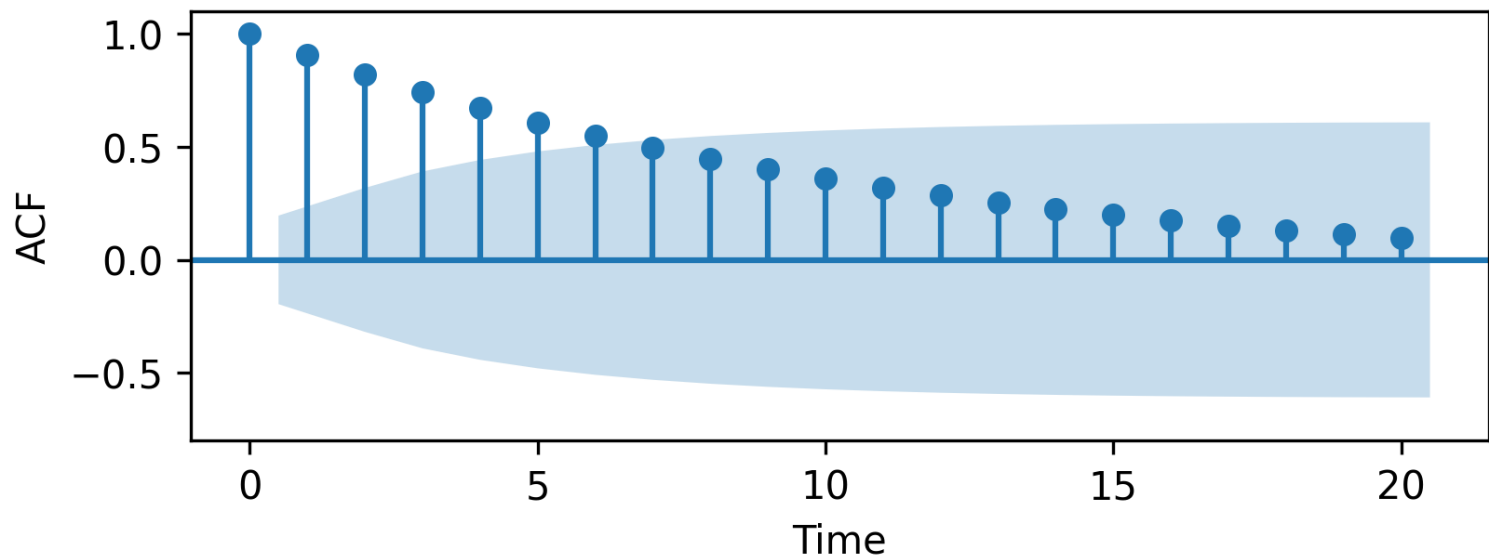
This is an example of exponential growth, though not the right way to model it

There are other models (and courses) for studying growth curves

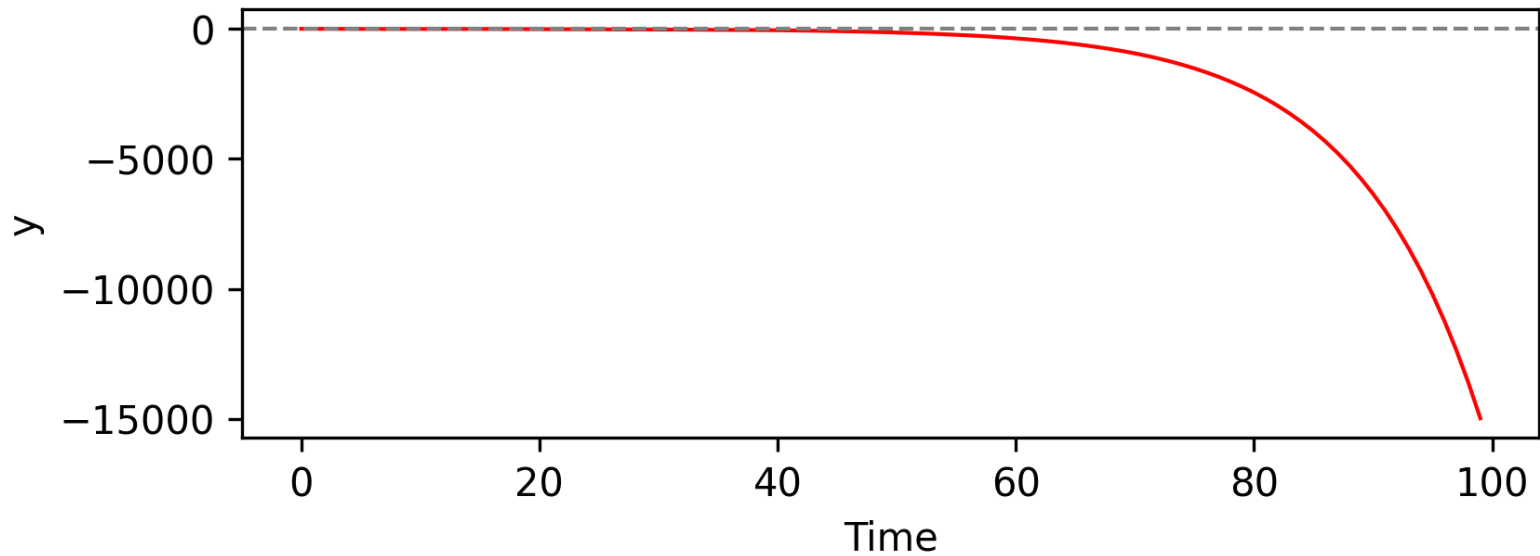
Simulated AR(1) process with $\phi_1 = 1.1$



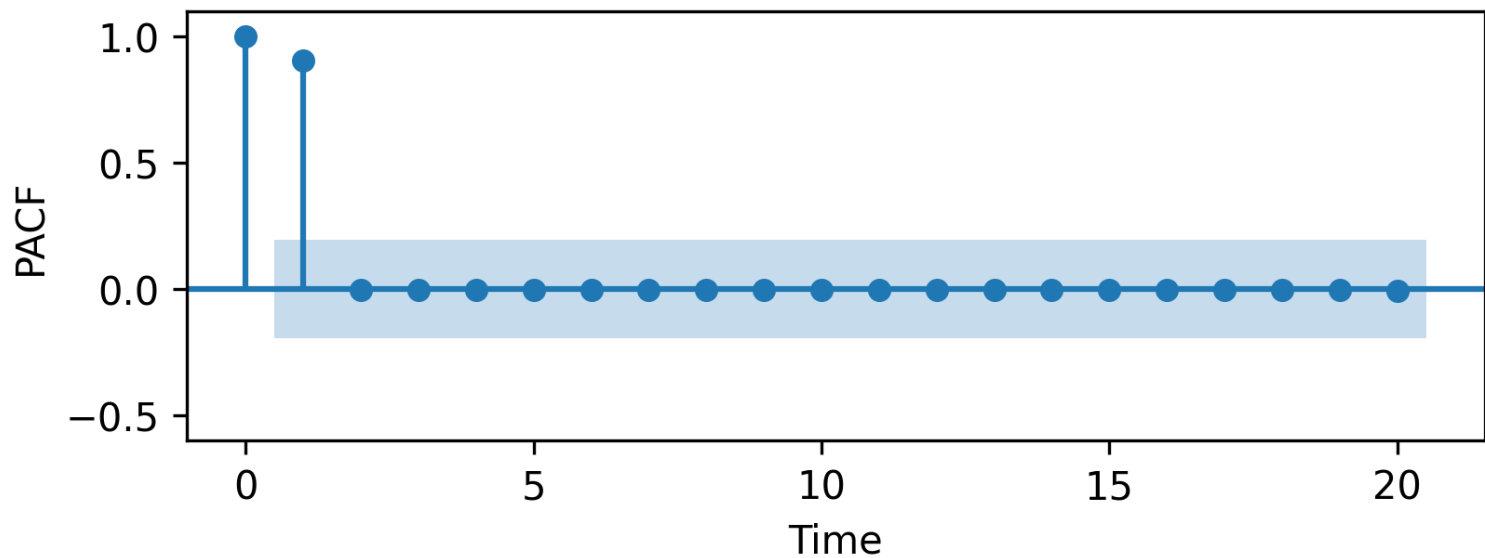
ACF of AR(1) process with $\phi_1 = 1.1$



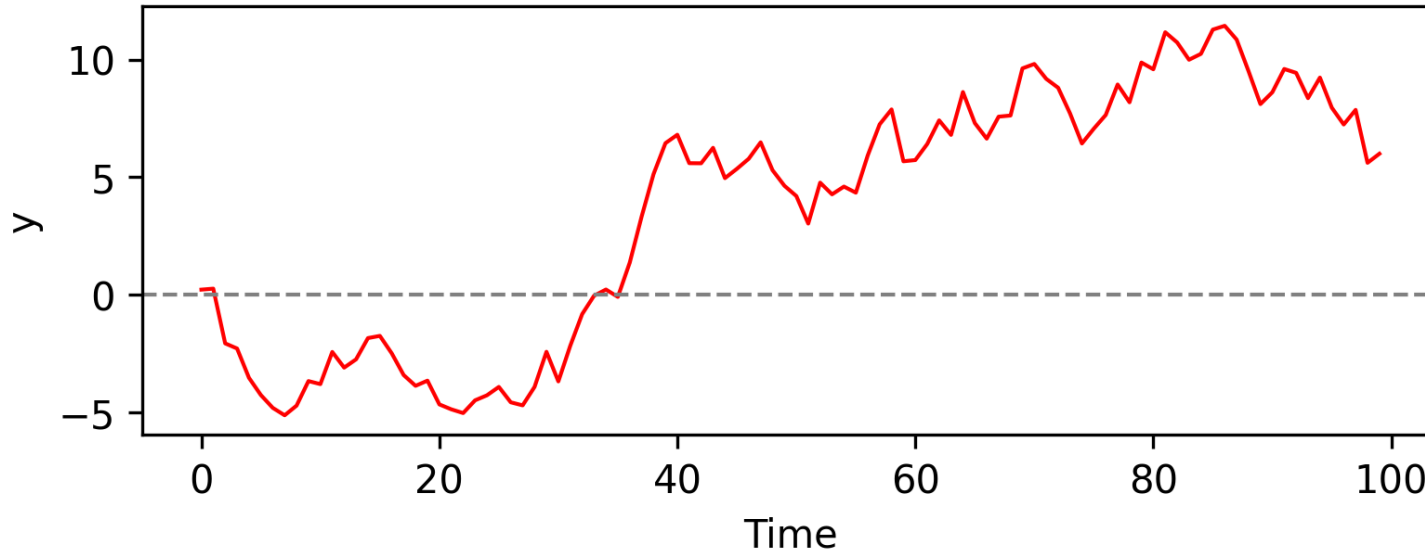
Simulated AR(1) process with $\phi_1 = 1.1$



PACF of AR(1) process with $\phi_1 = 1.1$



Simulated AR(1) process with $\phi_1 = 1$

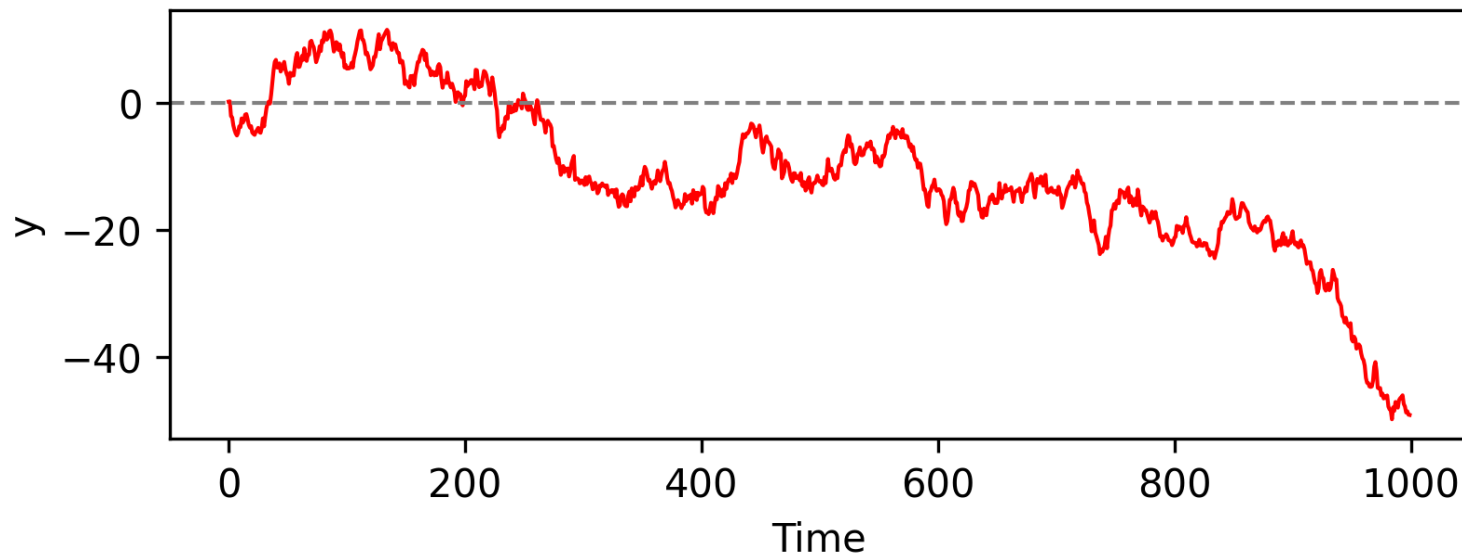


Is $\phi_1 = 1$ really different from $\phi_1 = 0.99$?

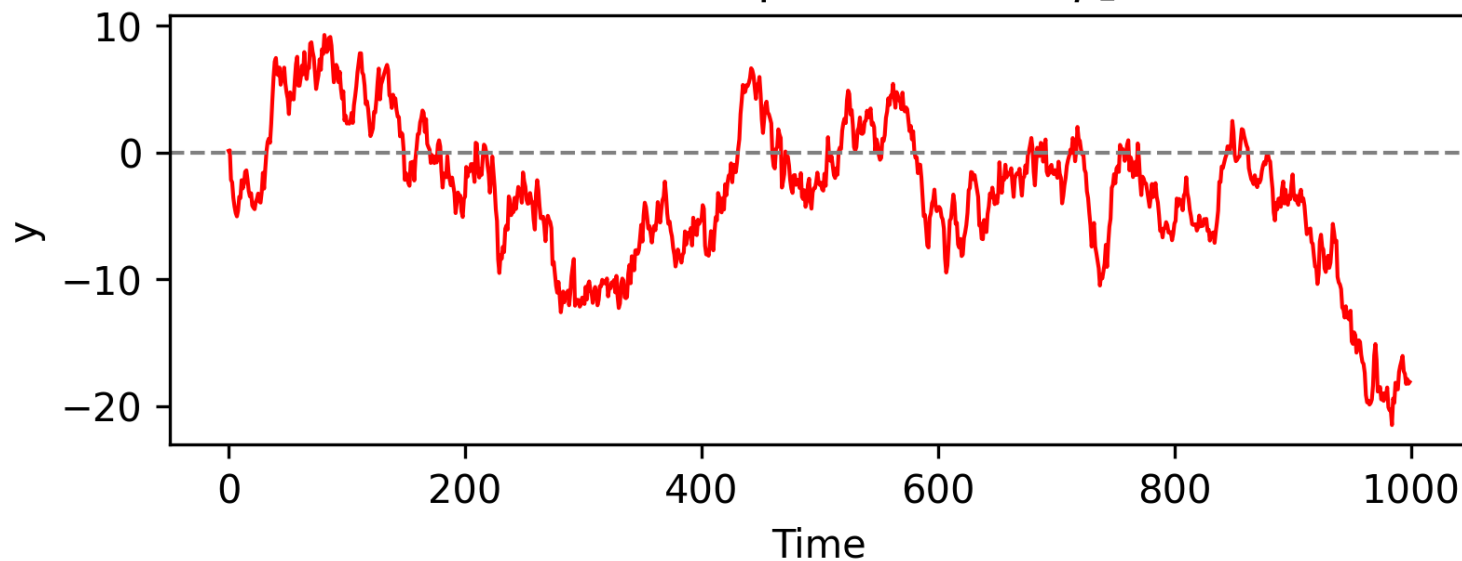
Will both eventually bounce around the mean?

Let's extend the sequences for $\phi_1 = 1$ and $\phi_1 = 0.99$

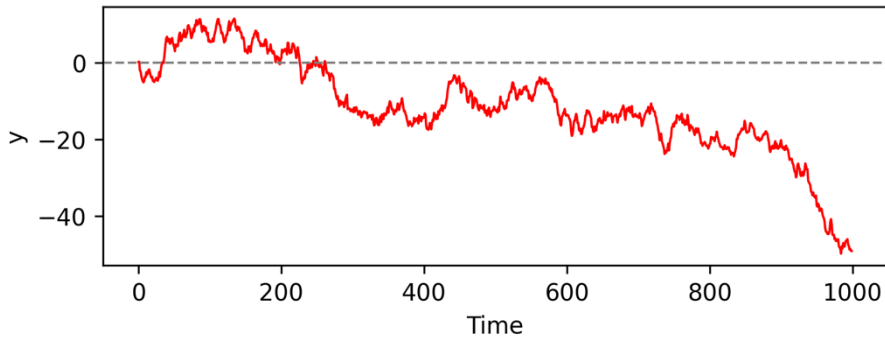
Simulated AR(1) process with $\phi_1 = 1$



Simulated AR(1) process with $\phi_1 = 0.99$



Simulated AR(1) process with $\phi_1 = 1$

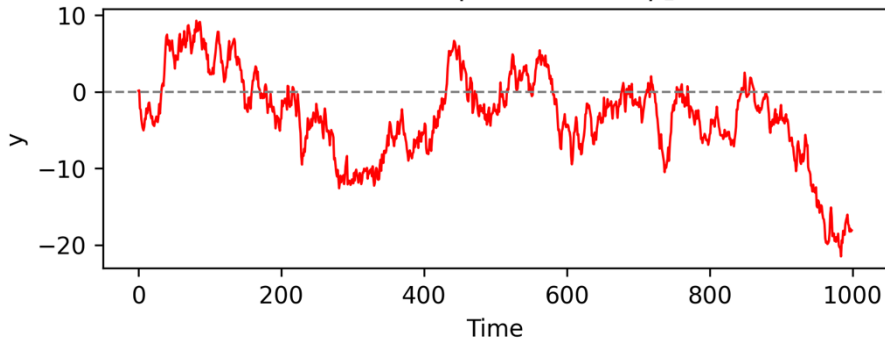


Major difference between $\phi_1 = 1$ and $\phi_1 = 0.99$

$$\phi_1 = 1$$

$\phi_1 = 1$ doesn't mean revert ever.
Only passes back through 0 by chance.

Simulated AR(1) process with $\phi_1 = 0.99$

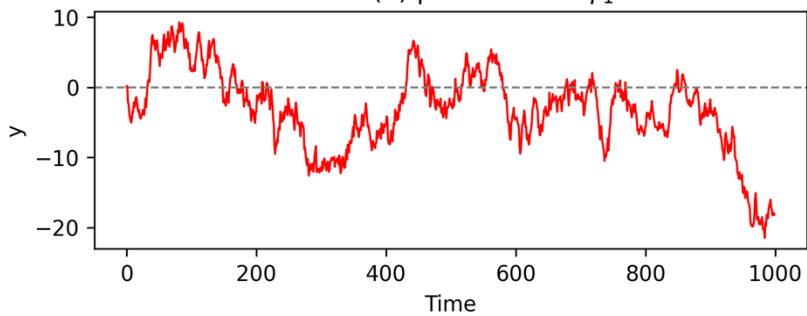


$$\phi_1 = 0.99$$

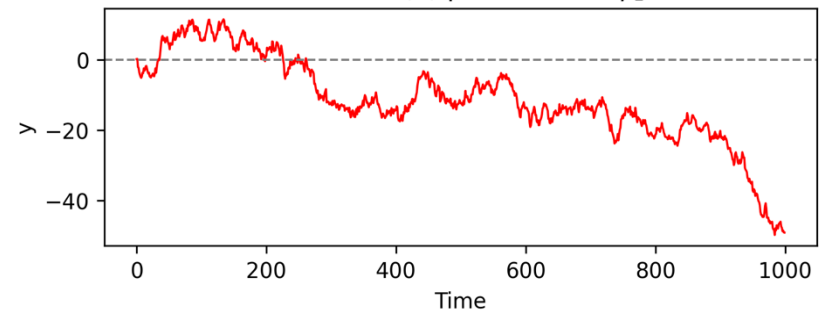
Note that it stays away from the mean
for a while but always comes back.

So does the $\phi_1 = 0.999$ sequence, and
the $\phi_1 = 0.9999$ sequence

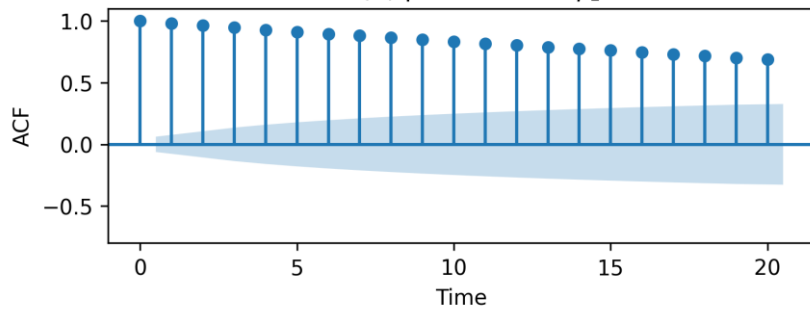
Simulated AR(1) process with $\phi_1 = 0.99$



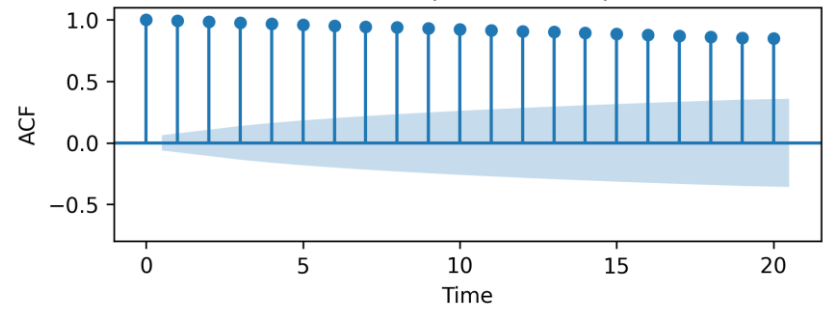
Simulated AR(1) process with $\phi_1 = 1$



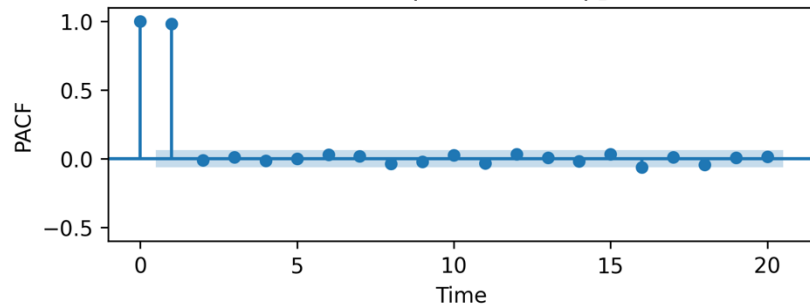
ACF of AR(1) process with $\phi_1 = 0.99$



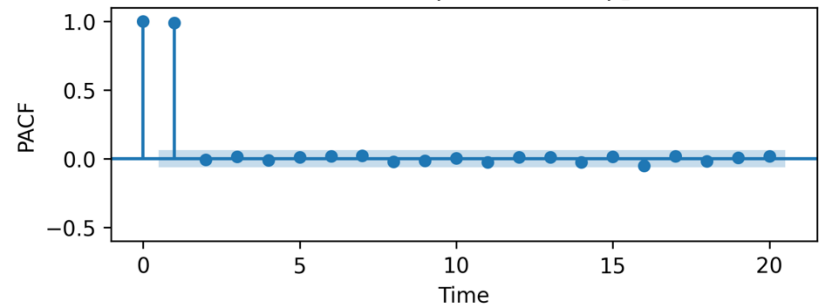
ACF of AR(1) process with $\phi_1 = 1$



PACF of AR(1) process with $\phi_1 = 0.99$



PACF of AR(1) process with $\phi_1 = 1$



$\phi_1 = 1$ and $\phi_1 = 0.99$ have almost identical ACF and PACF

Stationary Series

AR(1) processes with $|\phi_1| > 1$ lack three related properties which all AR(1) processes with smaller ϕ_1 possess:

1. Mean stationarity
2. Covariance stationarity
3. Ergodicity

Mean stationarity

A time series is mean *stationary* if its mean does not depend on t

Formally, $\mathbb{E}(y_t) = \mu$ for all t

Recall we are abstracting from deterministic trends and other covariates
As such, we might be working with a detrended series like

$$y_t - t\beta = 0.5y_{t-1} + \varepsilon_t$$

Although y_t trends upward deterministically before detrending,
the rest of the components of $y_t - t\beta$ are mean-stationary

Time series that are stationary after detrending are *trend stationary*

Covariance stationary

A time series is covariance stationary if neither the mean of y_t nor the covariance of y_t and y_{t-k} depend on t

The covariance may still depend on the length of time between two observations, k

Formally,

$$\text{cov}(y_t, y_{t-k}) = \mathbb{E}((y_t - \mu)(y_{t-k} - \mu)) = \gamma_k$$

for all t and k

Also known as “weak stationarity”

Today's Learning Outcomes

1. Understand what autoregressive process is
2. Understand ACF and PACF
3. Understand stationarity